

Project Genesis & Infrastructure Foundation

Lines of Code 500+	Tests CI Setup	Docker Running	DB Migrated
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Technical Architecture

- FastAPI backend with strict Pydantic v2 schemas.
- Postgres (via Docker) with Alembic migrations - essential for rollbacks.
- Redis for hot-path caching and real-time pub/sub.
- React + Vite frontend shell, prioritizing high-performance rendering.

Autopilot AI Logic

- The AI CANNOT act if the sensors (APIs) are flaky.
- This week focuses on defining the *interfaces* the AI will use.
- Centralized logging (structlog) is critical for post-mortem AI analysis.
- Environment variables must be loaded via typed `Settings`` class - avoiding runtime config errors.

Operational Playbook

- Run ``make dev`` to start the full stack (API + DB + Redis).
- Always verify the migration created by ``alembic revision --autogenerate``.
- Use ``pre-commit`` hooks to prevent committing secrets or lint errors.
- Set up GitHub Actions immediately to catch regressions early.

Risk & Contingency

- Risk: Database schema drift. Mitigation: CI check for missing migrations.
- Risk: Secrets in git. Mitigation: trufflehog scan in CI.
- Risk: Dependency hell. Mitigation: Lock files (poetry.lock or requirements.txt pinned).

Day-by-Day Schedule

Mon	Repo init, comprehensive .gitignore, Docker Compose setup.
Tue	FastAPI boilerplate, Health checks, Structured Logging.
Wed	Database Models (User, Trade), Alembic setup, first migration.
Thu	Redis integration, basic Auth (JWT) scaffolding.
Fri	CI/CD pipeline (GitHub Actions), Readme, Developer Onboarding doc.

Broker API Integration (Alpaca/Tradier)

Broker Connected	Latency <500ms	Errors Handled	Mock 100%
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Technical Architecture

- Broker Interface (Abstract Base Class) allows swapping providers later.
- Alpaca implementation for paper trading and market data.
- Tradier implementation for options chains (sandbox initially).
- Rate-limiter decorator ensuring we respect API quotas.

Autopilot AI Logic

- AI needs confirmable execution status (PENDING -> FILLED).
- Implement "blind" retry logic cautiously; prefer checking status first.
- MockBroker is critical: train the AI against a deterministic market simulator.
- Secrets management: Load keys from .env, never hardcode.

Operational Playbook

- Test connectivity using ``scripts/check_broker.py``.
- Monitor API usage headers to tune rate limiters.
- Use the Sandbox/Paper environment exclusively for now.
- Log full request/response bodies (sanitized) for debugging failures.

Risk & Contingency

- Risk: Infinite retry loops. Mitigation: Exponential backoff with jitter.
- Risk: API downtime. Mitigation: Circuit breaker pattern (3 failures -> open).
- Risk: Key leakage. Mitigation: Enforce env var injection only.

Day-by-Day Schedule

Mon	Define Broker ABC (buy, sell, get_position, get_account).
Tue	Implement Alpaca adapter (REST API wrapper).
Wed	Implement MockBroker for unit testing (critical!).
Thu	Integration tests running against Sandbox API.
Fri	Rate limiting & Retry middleware implementation.

Market Data Ingestion & Storage

Feed Live	DB Size Growing	Bars 1-Min	Gaps 0%
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Technical Architecture

- Asyncio fetcher for concurrent symbol data retrieval.
- TimescaleDB (or partitioned Postgres) for efficient time-series storage.
- Data normalization layer: ensure all vendors map to common schema.
- Gap detection worker: re-fetch missing bars automatically.

Autopilot AI Logic

- AI models fail on partial data (NaNs). Input sanitization is key.
- Store raw vendor responses if possible for audit/replay.
- Real-time WebSocket client for live price updates.
- Market schedule awareness (don't fetch when closed).

Operational Playbook

- Schedule cron jobs via APScheduler (or system cron) for EOD sync.
- Monitor disk usage; time-series data grows fast.
- Verify data consistency against a second source if possible.
- Use `pandas` for bulk localized timezone conversion.

Risk & Contingency

- Risk: Rate limit exhaustion during backfill. Mitigation: Throttled queue.
- Risk: Timezone confusion (UTC vs ET). Mitigation: Store UTC, display ET.
- Risk: Data corruption. Mitigation: Checksums or row counts vs vendor.

Day-by-Day Schedule

Mon	Schema design for OHLCV bars and Quotes.
Tue	Historical data backfill script (1 year, 1-min bars).
Wed	Real-time WebSocket client implementation.
Thu	Data normalization & Validation (check for spikes/zeros).
Fri	Gap detection service & auto-patching.

Frontend Dashboard Implementation

Pages 5	Load <1s	Updates Real-time	Theme Dark
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Technical Architecture

- React (Vite) + TypeScript for type-safe components.
- Tailwind CSS for rapid styling (Dark Mode default).
- Recharts or Lightweight-Charts for financial visualization.
- UseQuery for server state management (caching, polling).

Autopilot AI Logic

- The dashboard must show *intent*, not just results.
- Display "AI Reasoning" logs prominently (why did it buy?).
- Kill Switch button must be physical/prominent and instant.
- Visualize "Confidence Score" for every potential trade.

Operational Playbook

- Keep components small and functional.
- Use a shared layout for Navigation and Status Bar.
- Mock API responses for frontend development speed.
- Implement "Toast" notifications for critical alerts.

Risk & Contingency

- Risk: Stale data. Mitigation: WebSocket pushed updates or frequent polling.
- Risk: Complex state. Mitigation: Use typed contexts or Zustand stores.
- Risk: UI blocking. Mitigation: Offload heavy calc to web workers or backend.

Day-by-Day Schedule

Mon	Layout skeleton (Sidebar, Header, Main Content).
Tue	Dashboard Widget: Portfolio Summary & Active P&L.;
Wed	Data Grid: Open Positions with sorting/filtering.
Thu	Chart Component: TradingView-style interactive chart.
Fri	"System Status" panel (Broker connection, API health).

Options Engine Core Logic

Greeks Computed	Latency <10ms	Chain Full	IV Solved
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Technical Architecture

- Numpy-vectorized Black-Scholes solver.
- OptionChain object model.
- IV Root-finding (Newton-Raphson).
- Redis caching for processed chains.

Autopilot AI Logic

- AI trades volatility, not just price.
- Accurate Delta/Theta are inputs to the decision model.
- Chains are large; filter OTM strikes early to save compute.

Operational Playbook

- validate inputs ($T > 0$, $Vol > 0$).
- Handle "divide by zero" near expiration.
- Verify Greeks against a known source (e.g., broker).
- Cache results for 1 minute.

Risk & Contingency

- Risk: Model error. Mitigation: Unit tests against text-book examples.
- Risk: Stale IV. Mitigation: Recompute on price updates.
- Risk: Bad data. Mitigation: Filter bids=0 or abnormal spreads.

Day-by-Day Schedule

Mon	Black-Scholes implementation & tests.
Tue	Implied Volatility solver.
Wed	Greeks calculator (Delta, Gamma, Vega, Theta).
Thu	Option Chain data structure & parser.
Fri	Performance optimization (vectorization).

LLM Integration (Groq & Gemini)

LLMs 2	Cost Tracked	Latency <2s	Fallback Yes
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Technical Architecture

- LangChain or LiteLLM wrapper for Unified Interface.
- Groq (Llama 3) for speed (Decision).
- Gemini Pro 1.5 for context (Analysis).
- Prompt registry (versioned text files).

Autopilot AI Logic

- The LLM is a reasoning engine, not a database.
- Provide structured context (JSON) in the prompt.
- Force structured output (JSON mode) for parsing.
- Dual-LLM consensus: if they disagree, don't trade.

Operational Playbook

- Log every prompt and response to DB.
- Monitor token usage costs.
- Implement retry logic for API timeouts.
- Use strict system prompts to define persona.

Risk & Contingency

- Risk: Hallucination. Mitigation: Grounding in provided data only.
- Risk: Format error. Mitigation: Pydantic validation of output.
- Risk: Rate limits. Mitigation: Queueing system.

Day-by-Day Schedule

Mon	API Client wrappers & Auth.
Tue	Prompt Engineering: "Market Analyst Persona".
Wed	Structured Output parsers (Pydantic).
Thu	Dual-LLM consensus logic.
Fri	Cost tracking & logging middleware.

The Auto-Scheduler (Cron & Events)

Jobs Running	Drift <1s	Missed 0	Holiday Aware
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Technical Architecture

- APScheduler (AsyncIO) for task orchestration.
- Market Calendar integration (NYSE holidays).
- Event bus for signal-driven triggers.
- Distributed locking (Redis) to prevent double-runs.

Autopilot AI Logic

- AI needs to wake up before market open (Pre-market scan).
- Intraday checks every 5/15 minutes.
- Post-market cleanup and journaling.
- Don't run on weekends (unless crypto).

Operational Playbook

- Check `trading\_calendars` library for holidays.
- Monitor job heartbeat.
- Handle daylight savings time logic automatically (pytz).
- Graceful shutdown signal handling.

Risk & Contingency

- Risk: Double execution. Mitigation: Redis locks.
- Risk: Silent failure. Mitigation: Dead Man's Switch monitoring.
- Risk: Server clock drift. Mitigation: NTP sync.

Day-by-Day Schedule

Mon	APScheduler setup & config.
Tue	Market Calendar integration.
Wed	Job definitions (Scan, Manage, Report).
Thu	Redis locking mechanism.
Fri	Admin UI for Manual Job Triggering.

Risk Management & The Kill Switch

Gates Active	Latency 0ms	Manual Override	Max Loss Enforced
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Technical Architecture

- Risk Gate middleware (checks every order).
- Portfolio-level limits (Max DD, Max Alloc).
- Kill Switch service (panic button).
- Position Sizer (Kelly / Fixed Fractional).

Autopilot AI Logic

- AI is aggressive; Risk Manager is conservative.
- Hard limits that the LLM cannot override.
- Daily Loss Limit: Stop trading if down X%.
- Correlaton check: Don't stack correlated risks.

Operational Playbook

- Test the Kill Switch regularly (in paper).
- Alert immediately on Risk Rejection.
- Review rejected trades to tune sensitivity.
- Keep risk params in a hot-reloadable config.

Risk & Contingency

- Risk: Latency preventing exit. Mitigation: Local stop-loss orders.
- Risk: Config error. Mitigation: Sanity bounds (e.g., max 5%).
- Risk: Emotional override. Mitigation: Log all overrides.

Day-by-Day Schedule

Mon	Position Sizing logic implementation.
Tue	Portfolio-level Hard Limits (Max DD).
Wed	Order-level Gate (Max Size, Min Liq).
Thu	Kill Switch & Liquidation logic.
Fri	Integration tests: Try to break the rules.

Strategy Interface & Strategy #1

Strategy 1	Signals Generated	Profit Unknown	Code Modular
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Technical Architecture

- Strategy Abstract Base Class.
- Signal Generator (Entry).
- Management Logic (Adjust/Exit).
- Configuration schema per strategy.

Autopilot AI Logic

- Strategies are plugins to the autopilot.
- Strategy #1: Iron Condor on Indices (SPY/IWM).
- Logic: High IV Rank -> Sell Wings.
- Management: Close at 50% profit or 2x loss.

Operational Playbook

- Backtest the specific logic first.
- Start with small size (1 contract).
- Log "Why" the strategy fired.
- Parameterize everything (deltas, DTE, etc.).

Risk & Contingency

- Risk: Legging out risk. Mitigation: Use complex orders (Iron Condor).
- Risk: Assignment. Mitigation: Close before expiration (21 DTE).
- Risk: Gap risk. Mitigation: Defined risk spreads.

Day-by-Day Schedule

Mon	Definition of Strategy Interface.
Tue	Signal Generation Logic (Entry).
Wed	Exit/Adjustment Logic.
Thu	Iron Condor implementation.
Fri	Dry-run testing of Strategy #1.

Notification System (Discord/Slack/SMS)

Channels Discord	Latency <1s	Rich Embeds	Actions Links
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Technical Architecture

- Notification Service (Async).
- Discord Webhook integration.
- Template engine for messages.
- Priority levels (Info vs Critical).

Autopilot AI Logic

- Notify on: Trade Entry/Exit, Risk Alert, Daily Summary.
- Don't spam: Digest frequent low-priority events.
- Critical alerts must wake you up (PagerDuty/SMS).
- Include charts/screenshots if possible.

Operational Playbook

- Use distinct channels for #trades, #errors, #logs.
- Verify webhook reliability.
- Include "Kill Switch" link in critical alerts.
- Sanitize messages (no API keys).

Risk & Contingency

- Risk: API Rate limits. Mitigation: Batching/Queue.
- Risk: Missed alert. Mitigation: Multiple channels.
- Risk: Noise fatigue. Mitigation: Configurable verbosity.

Day-by-Day Schedule

Mon	Notification Service architecture.
Tue	Discord Webhook & Embed builder.
Wed	Event listeners (Trade -> Notify).
Thu	Daily Briefing generator.
Fri	SMS/Email fallback integration.

Backtesting Engine v1

Speed Fast	Accuracy High	Reports PDF	Engine Event
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Technical Architecture

- Event-driven backtester (matches live engine).
- Historical Data Feed adapter.
- Order Matching Simulator (Slippage/Comm).
- Performance Reporter (Sharpe/Sortino).

Autopilot AI Logic

- Test Strategy #1 (IC) on last year's data.
- Verify matching logic against known outcomes.
- Simulate latency and partial fills.
- Run parameter sweeps (Optimization).

Operational Playbook

- Don't overfit. Walk-forward testing is next.
- Visualize the equity curve.
- Check for "look-ahead bias" in data access.
- Save results to DB for comparison.

Risk & Contingency

- Risk: Data snooping. Mitigation: Strict timestamp enforcement.
- Risk: Unrealistic fills. Mitigation: Pessimistic slippage model.
- Risk: Curve fitting. Mitigation: Out-of-sample validation.

Day-by-Day Schedule

Mon	Event-Loop Engine skeleton.
Tue	Historical Feed & Matcher.
Wed	Performance Metrics calc.
Thu	Reporting module (Graphs).
Fri	Run IC Strategy Backtest 2023.

Database Optimization & Maintenance

Query <50ms	Index Optimized	Backup Daily	Size Managed
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Technical Architecture

- Postgres Indexing strategy (B-Tree, BRIN).
- Partitioning for time-series tables.
- Vacuum & Analyze schedules.
- Connection Pooling (pgbouncer).

Autopilot AI Logic

- Slow DB = Slow Decision = Slippage.
- Optimize the "Hot Path" queries.
- Archive old data (cold storage).
- Monitor connection usage.

Operational Playbook

- Run `EXPLAIN ANALYZE` on slow queries.
- Set up automated backups (WAL-G or pg_dump).
- Monitor lock contention.
- Clean up ephemeral logs.

Risk & Contingency

- Risk: Disk full. Mitigation: Alerts at 80%.
- Risk: Query Timeout. Mitigation: Statement timeouts.
- Risk: Data Loss. Mitigation: RAID + Offsite backups.

Day-by-Day Schedule

Mon	Query Profiling & Analysis.
Tue	Index creation & tuning.
Wed	Table Partitioning (Monthly).
Thu	Backup & Restore scripts.
Fri	Connection Pool tuning.

Q1 Wrap-Up: Integration & Dry Run

Uptime 99%	DryRun Success	Logs Clear	Ready Q2
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Technical Architecture

- System Integration Test.
- Full Environment Swing check.
- Documentation Review.
- Load Testing.

Autopilot AI Logic

- Run the full autopilot cycle in "Paper" mode.
- Verify end-to-end flow: Scan -> Score -> Trade -> Report.
- Check handling of edge cases (network down, etc.).
- Review "Confidence" of AI decisions.

Operational Playbook

- Freeze code for Q1 release.
- Conduct a "Pre-Mortem" meeting.
- Clean up tech debt.
- Plan Q2 features.

Risk & Contingency

- Risk: Integration failure. Mitigation: Fix immediately.
- Risk: Complexity creep. Mitigation: Refactor complex modules.
- Risk: Burnout. Mitigation: Celebrate the milestone!

Day-by-Day Schedule

Mon	Full System Integration Test.
Tue	Fix bugs & Polish logs.
Wed	Load Test (simulate high traffic).
Thu	Documentation & Runbooks update.
Fri	Q1 Release & Team Celebration.

Walk-Forward Optimization (The Anti-Overfit)

Windows 12	OOS Ratio >0.8	Params Stable	Overfit None
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Technical Architecture

- Window Splitter: In-Sample (IS) vs Out-of-Sample (OOS).
- Parameter Grid Search (Optuna/Hyperopt).
- Stability Metric: IS/OOS Performance Ratio.
- WFO Report Generator.

Autopilot AI Logic

- AI must select parameters based on OOS performance ONLY.
- If parameters drift wildly between windows, the strategy is broken.
- Automated "Regime Detection": Does strategy fail in 2022 (bear)?
- Reject any strategy with Sharpe < 1.0 in OOS.

Operational Playbook

- Define 12 rolling windows (e.g., 3 months train, 1 month test).
- Run optimization on High Performance Cluster (or 64-core AWS instance).
- Don't optimize everything—pick 2-3 key params (e.g., specific deltas).
- Visualize parameter stability heatmaps.

Risk & Contingency

- Risk: Look-ahead bias in WFO. Mitigation: Strict index slicing.
- Risk: Over-optimization. Mitigation: Deflated Sharpe Ratio (DSR).
- Risk: Survivor bias. Mitigation: Include delisted symbols.

Day-by-Day Schedule

Mon	Window Splitter implementation.
Tue	Integration with Optuna for parameter search.
Wed	WFO Engine logic (Train -> Test -> Roll).
Thu	Aggregation of OOS equity curves.
Fri	Analyze Strategy #1 with WFO. Pass/Fail?

Monte Carlo Simulations & VaR

Sims 10,000	VaR 95 Calc	Ruin <1%	Confidence High
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Technical Architecture

- Monte Carlo Engine: Resampling trade history.
- Sequence Risk Analyzer.
- Value at Risk (VaR) calculator (Historical + Parametric).
- Equity Curve Simulator.

Autopilot AI Logic

- AI uses MC to size positions: "What size keeps risk of ruin < 5%?"
- Simulate "Black Swan" events (3-sigma moves).
- Adjust leverage based on MC Ruin probability.
- Reject strategies with Fat Tails (excessive kurtosis).

Operational Playbook

- Run 10,000 simulations per strategy.
- Visualize the "Cone of Uncertainty".
- Check max drawdown duration—can you survive 6 months underwater?
- Calculate "Risk of Ruin" given current capital.

Risk & Contingency

- Risk: Assuming normal distribution. Mitigation: Use bootstrap resampling.
- Risk: Correlation ignorance. Mitigation: Shuffle days, not just trades.
- Risk: False confidence. Mitigation: Use worst-case outcome.

Day-by-Day Schedule

Mon	Monte Carlo Engine (Numpy vectorized).
Tue	VaR (Value at Risk) implementation.
Wed	Sequence Risk visualization.
Thu	Integration with Position Sizer.
Fri	Stress Test Report for Q1 Strategies.

Portfolio Greeks Dashboard

B.Delta Neutral	Vega Managed	Theta Positive	View Live
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Technical Architecture

- Greek Aggregator: Summing Greeks across all positions.
- Beta-Weighting Engine (SPY as benchmark).
- Real-time Header Components (SPY Delta, Port Theta).
- Drill-down view per symbol.

Autopilot AI Logic

- AI Goal: Maintain Delta Neutrality (+/- 50 SPY deltas).
- Auto-Hedge: If Delta > 100, buy Puts or sell Calls.
- Theta Target: Target 0.1% daily theta decay.
- Vega limit: Don't exceed X% exposure to 1% IV expansion.

Operational Playbook

- Beta-weight individual stock deltas to SPY.
- Refresh Greeks every 1-5 minutes (computationally expensive).
- Alert if Gamma risk becomes too high (0 DTE).
- Visualize "Greeks over Time" to see stability.

Risk & Contingency

- Risk: Model mismatch. Mitigation: Re-calc IV live.
- Risk: Data lag. Mitigation: Timestamp check on quotes.
- Risk: Currency risk. Mitigation: Ignore for now (US-only).

Day-by-Day Schedule

Mon	Beta-weighting logic (Linear Regression of returns).
Tue	Backend Aggregation Service.
Wed	Frontend Greeks Dashboard (Bar charts).
Thu	Auto-Hedge Logic (Theoretical).
Fri	Alerting on Greek Thresholds.

Smart Execution Algorithms

Slippage <0.05	Fills Improved	Algo TWAP	Limit Adaptive
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Technical Architecture

- Execution Router (Smart Order Router logic).
- Pegged Orders (Mid-Price, NBBO offset).
- TWAP (Time Weighted Average Price) for large orders.
- Chase Logic: Limit order -> Wait -> Cancel/Replace.

Autopilot AI Logic

- Never use Market Orders on Options (wide spreads).
- Start at Mid-Price. Walk limit towards Ask (for buy) every 5s.
- Detect "Iceberg" orders or hidden liquidity.
- Cancel if price moves away too fast (Chase limit).

Operational Playbook

- Measure slippage on every trade (Exec Price vs Arrival Mid).
- Tune walk aggression based on volatility.
- Handle partial fills logic (accumulate position).
- Video replay of execution (log tick data during fill).

Risk & Contingency

- Risk: Runaway algorithm. Mitigation: Max order count limit.
- Risk: Stuck orders. Mitigation: TTL (Time To Live).
- Risk: API Reject. Mitigation: Handle bad tick size/price.

Day-by-Day Schedule

Mon	Adaptive Limit Order logic (Mid-Market Peg).
Tue	Chase Logic (Wait & Move).
Wed	TWAP implementation for > 10 contracts.
Thu	Slippage Analytics & Logging.
Fri	Stress test with MockBroker (simulating delays).

Automated Reporting Engine

PDF
Generated

Email
Sent

P&L;
Accurate

Charts
Embedded

Technical Architecture

- ReportLab (PDF generation) automation.
- Jinja2 Templates for HTML Email summaries.
- SMTP / SendGrid integration.
- Equity Curve plotter (Matplotlib/Seaborn).

Autopilot AI Logic

- AI generates a "Daily Commentary" (LLM based) on performance.
- Explain WHY trades were taken in plain English.
- Highlight "Near Misses" (trades filtered by Risk).
- Suggest parameter tweaks based on recent market regime.

Operational Playbook

- Scheduled job: 4:15 PM ET (Market Close).
- Include: Daily P&L;, YTD, Open Positions, Greeks.
- Archive reports to S3 or local disk.
- Add CSV export for Excel analysis.

Risk & Contingency

- Risk: Email failure. Mitigation: Retry & logging.
- Risk: Sensitive data. Mitigation: Encrypt attachments or use secure portal links.
- Risk: P&L; discrepancy. Mitigation: Reconcile with Broker daily.

Day-by-Day Schedule

Mon	PDF Template design (ReportLab).
Tue	Data aggregation for Daily Report.
Wed	LLM Commentary generation ("Today was choppy...").
Thu	Email delivery service.
Fri	Reconciliation Module (Internal vs Broker).

Dynamic Universe Management

Symbols Liquid	Update Daily	Bad Removed	Sectors Balanced
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Technical Architecture

- Universe Scanner (IV Rank, Liquidity, Volume).
- Blacklist Manager (Earnings, FDA approvals, Lawsuits).
- Sector Balancing logic.
- ETF constituents loader (holdings of SPY/QQQ).

■ Autopilot AI Logic

- AI decides WHAT to trade based on regime.
- Low Volatility? Scan for Calendars.
- High Volatility? Scan for Iron Condors/Credit Spreads.
- Auto-remove symbols with wide spreads or low volume.

■■ Operational Playbook

- Daily Pre-market scan (8:00 AM).
- Filter: Avg Vol > 1M shares, Option Vol > 5k.
- Maintain a "Watchlist" vs "Trade List".
- Check for "Hard to Borrow" if shorting stock.

■■ Risk & Contingency

- Risk: Bad data ticks. Mitigation: Outlier filter.
- Risk: MEME stock explosion. Mitigation: IV Rank & HV checks.
- Risk: Concentration. Mitigation: Max 2 positions per Sector.

■ Day-by-Day Schedule

Mon	Scanner Criteria Definition.
Tue	Pipeline implementation (Source -> Filter -> DB).
Wed	Blacklist logic (Earnings dates).
Thu	Sector Diversification logic.
Fri	Automated Universe Roll (Weekly update).

Test Suite Expansion (80% Coverage)

Cov >80%	Unit Pass	Integ Pass	Flaky 0
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Technical Architecture

- Pytest plugins (cov, xdist, randomly).
- Integration Test Environment (Dockerized).
- Property-based testing (Hypothesis library).
- Mutation Testing (mutmut) - finding gaps in tests.

Autopilot AI Logic

- Unit test the *Prompts*: Do they return valid JSON?
- Regression test the *Decision Engine*: Replay past market data.
- Fuzz testing: Send garbage data to API endpoints.
- Mocking external services is mandatory for speed.

Operational Playbook

- Enforce coverage thresholds in CI.
- Refactor "God Classes" into smaller testable units.
- Document "How to Write Tests" for future self.
- Fix all distinct TODOs in code.

Risk & Contingency

- Risk: False confidence. Mitigation: Mutation testing.
- Risk: Slow CI. Mitigation: Parallel execution (xdist).
- Risk: Flaky tests. Mitigation: Deterministic seed.

Day-by-Day Schedule

Mon	Coverage analysis & gap identification.
Tue	Service Layer Unit Tests.
Wed	API Integration Tests (Happy/Sad paths).
Thu	Hypothesis (Property-based) testing for math.
Fri	CI Pipeline optimization & Badge generation.

Volatility Surface Modeling

Smile Modeled	Skew Tracked	Arb Scanned	3D Chart
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Technical Architecture

- Volatility Surface interpolation (Cubic Spline).
- Skew Calculator (Put vs Call IV).
- Term Structure Analysis (Contango/Backwardation).
- 3D Surface Visualization helper.

Autopilot AI Logic

- AI checks Skew: If Puts are expensive, sell Puts (Bull Put Spread).
- AI checks Term Structure: If Backwardation, sell front-month premium.
- Detect "Kinks" in the surface (potential mispricing).
- Normalizing IV across different expiries.

Operational Playbook

- Store surface snapshots daily.
- Use consistent "Fixed Strike" or "Fixed Delta" grid.
- Visualize Surface changes over time.
- Feed Skew Metrics into Entry Logic.

Risk & Contingency

- Risk: Interpolation artifacts. Mitigation: SVI param model.
- Risk: Illiquid strikes. Mitigation: Filter wide spreads.
- Risk: Data volume. Mitigation: Store only fitted parameters.

Day-by-Day Schedule

Mon	Volatility Skew calc (25d Put / 25d Call).
Tue	Term Structure signal (M1 vs M2).
Wed	Surface Interpolation logic.
Thu	Integration into Strategy Selection.
Fri	Visualization page.

Earnings & Event Manager

Calendar Synced	Filter Active	Exits Timed	Safe Closes
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Technical Architecture

- Earnings Calendar API (AlphaVantage/Tradier).
- Event Scraper (Fed Minutes, CPI release).
- Liquidation Scheduler (Close before Event).
- Volatility Crush predictor.

Operational Playbook

- Sync calendar weekly.
- Alert on "Approaching Earnings" for portfolio holdings.
- Check Fed Calendar for FOMC days (No entries 1hr before).
- Visualize events on the Chart.

Autopilot AI Logic

- HARD RULE: No open positions through Earnings for non-binary specs.
- AI scans for High IV *without* upcoming earnings (Anomaly).
- Close positions 2 days before earnings if not intended as a gamble.
- Tag trades as "Earnings Play" if intended.

Risk & Contingency

- Risk: Date change. Mitigation: Daily re-sync.
- Risk: Data missing. Mitigation: Fallback provider.
- Risk: Holding through via oversight. Mitigation: Auto-close forced.

Day-by-Day Schedule

Mon	Earnings Data Feed integration.
Tue	Macro Event Feed (CPI, FOMC).
Wed	Filter Logic: "Is Safe To Trade?".
Thu	Auto-Close Scheduler.
Fri	Volatility Crush analysis logic.

Advanced Alerting (PagerDuty/Twilio)

SMS
Works

Call
Works

Escalate
Yes

Sleep
Soundly

Technical Architecture

- PagerDuty / Twilio / OpsGenie integration.
- Health Check Daemon (distinct from main app).
- Log Monitor (Error -> Alert).
- Heartbeat Monitor (Dead man's switch).

Autopilot AI Logic

- Critical Errors (API Auth fail, DB Correlation drift) = Phone Call.
- Wanrning (Slippage high) = Slack/Discord.
- Info (Trade placed) = Log/Discord.
- AI analyzes logs to suggest "Silence Rules" for noise.

Operational Playbook

- Set up "On Call" schedule (even if it's just you).
- Test "Wake Me Up" functionality.
- Monitor "System Resources" (CPU/RAM/Disk).
- Price Alerts (Portfolio Stop Loss).

Risk & Contingency

- Risk: Alert Fatigue. Mitigation: Smart grouping.
- Risk: SMS cost. Mitigation: Rate limit specific errors.
- Risk: Monitor down. Mitigation: External monitoring (UptimeRobot).

Day-by-Day Schedule

Mon	Critical vs Warning classification.
Tue	Twilio / SMS integration.
Wed	External Heartbeat Check.
Thu	Resource Monitoring (Psutil).
Fri	Fire Drill: Simulate outage.

Data Integrity & Sanitation

BadTick 0	Spike Filtered	Gap Filled	Clean Yes
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Technical Architecture

- Outlier Detection Middleware (Z-Score > 5).
- Bad Tick filter (Price 0 or negative).
- Quote/Trade Cross check (Trade outside NBBO).
- Data Replay validation.

Autopilot AI Logic

- AI ignores price spikes of > X% in 1 second.
- Consistency Check: Last Price within Bid/Ask.
- Volume Check: Don't trade on 1-share prints.
- Fallback to secondary feed if primary goes dark.

Operational Playbook

- Clean historical DB of known bad ticks.
- Implement "Quarantine" table for suspect data.
- Visualize "Filtered Ticks" to tune sensitivity.
- Log all discarded data.

Risk & Contingency

- Risk: Filtering real crash. Mitigation: check broad market correlation.
- Risk: Latency. Mitigation: Optimized numpy filters.
- Risk: Vendor error. Mitigation: Report to vendor.

Day-by-Day Schedule

Mon	Outlier Detection Logic.
Tue	Quote/Trade Consistency Check.
Wed	Gap Filling Logic update.
Thu	Bad Tick Database Cleaner.
Fri	Feed Reliability metrics.

UI Polish & User Experience

Lighthouse 100	FPS 60	Mobile View	Theme Pro
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Technical Architecture

- Framer Motion for layout transitions.
- React Tooltip for dense data explanation.
- Responsive Grid Layout (Mobile/Desktop).
- Optimized Re-renders (React.memo).

Autopilot AI Logic

- Visualize the "Thinking Process" (Step 1..2..3).
- Show "Confidence Interval" bands on charts.
- Animated "Pulse" when scanning.
- Clear "Why I did this" tooltips on trade history.

Operational Playbook

- Dark Mode refinement (Contrast ratios).
- Keyboard shortcuts (Exec, Cancel).
- Loading Skeletons data fetching.
- Error Boundaries with recovery actions.

Risk & Contingency

- Risk: UI lag. Mitigation: Virtualization for large lists.
- Risk: Mobile touch. Mitigation: Touch targets > 44px.
- Risk: Info Overload. Mitigation: Progressive disclosure.

Day-by-Day Schedule

Mon	Layout Polish & Spacing.
Tue	Animations & Transitions.
Wed	Tooltips & Help Text.
Thu	Mobile Responsiveness.
Fri	Performance Profiling (React DevTools).

V1.0 Launch Candidate

Bug 0	Docs Complete	Release Tagged	Ready Yes
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Technical Architecture

- Release Branching (Git Flow).
- Docker Image Tagging (v1.0.0).
- Deployment Script (Ansible/Terraform).
- Final Smoke Test Suite.

Autopilot AI Logic

- Full 1-week Paper Trading Burn-in without intervention.
- Verify ALL logic paths one last time.
- Lock Model versions (Prompts).
- Review P&L; attribution logic.

Operational Playbook

- Code Freeze: No new features.
- Update README, API Docs, Runbooks.
- Backup everything.
- Reset Paper Account for clean start.

Risk & Contingency

- Risk: Last minute change. Mitigation: REJECT PRs.
- Risk: Env drift. Mitigation: Fresh clone & build.
- Risk: Panic. Mitigation: Checklist.

Day-by-Day Schedule

Mon	Code Freeze & Dependency Lock.
Tue	Full Regression Test Run.
Wed	Documentation Polish.
Thu	Burn-in Monitor.
Fri	V1.0.0 Release Tag & Party.

Advanced LLM Prompting (Chain-of-Thought)

Prompts v2	Accuracy +20%	Reasoning Rich	Cost Optimized
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Technical Architecture

- Prompt Registry v2 (Jinja2 templates).
- Few-Shot examples injection (Dynamic RAG).
- Chain-of-Thought (CoT) structure: "Let's think step by step".
- Output Validator (Guardrails).

Autopilot AI Logic

- AI must explain "Why NOT" to take a trade (Filter logic).
- Inject recent similar successful trades as context.
- Use "Critique" step: LLM reviews its own plan before acting.
- Structured JSON schema enforcement is mandatory.

Operational Playbook

- A/B Test Prompts (Version A vs Version B).
- Log "Reasoning Traces" for manual review.
- Tune "Temperature" per task (0.1 for execution, 0.7 for ideation).
- Monitor Token Window limits.

Risk & Contingency

- Risk: Hallucination of market rules. Mitigation: Hard-coded constraints.
- Risk: Context Overflow. Mitigation: Summarization of history.
- Risk: Prompt Injection (Internal). Mitigation: Sanitize inputs.

Day-by-Day Schedule

Mon	CoT Prompt Design ("State observation -> Hypothesis -> Plan").
Tue	Dynamic Context Injection (RAG-lite).
Wed	Self-Correction Loop implementation.
Thu	A/B Testing Framework for Prompts.
Fri	Analysis of Improved Decision Quality.

Portfolio Correlation Matrix

Corr
<0.5

Heatmap
Live

Cluster
View

Risk
Lowered

Technical Architecture

- Correlation Matrix Calculation (Rolling 30-day).
- Hierarchical Risk Parity (HRP) clustering.
- Dendrogram Visualization.
- New Risk Gate: "Too Correlated".

Autopilot AI Logic

- Before Entry: Check correlation with existing Open Positions.
- If $\text{Corr}(\text{New}, \text{Existing}) > 0.7$, REJECT or reduce size.
- Aim for "Orthogonal Bets" (Uncorrelated returns).
- Detect "Risk On/Off" clusters.

Operational Playbook

- Re-calculate Matrix daily (Compute heavy).
- Visualize Heatmap in Dashboard.
- Alert on "Sector Concentration" (e.g., too much Tech).
- Use ETF proxies for sector correlation.

Risk & Contingency

- Risk: Spurious correlation. Mitigation: Longer lookback window.
- Risk: Regime change. Mitigation: Exponential weighting.
- Risk: Data gaps. Mitigation: Pairwise complete obs.

Day-by-Day Schedule

Mon	Rolling Correlation Engine (Pandas).
Tue	Integration with Risk Gate.
Wed	Heatmap Visualization (D3/Recharts).
Thu	Cluster Analysis (Scikit-Learn).
Fri	Backtest: Correlation Filter impact.

Advanced Exit Logic (Scaling Out)

Profits Secured	Runners Held	Loss Cut	Auto Trailed
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Technical Architecture

- Multi-Leg Exit Manager.
- Trailing Stop Logic (ATR based).
- Scale-Out Trigger (Limit orders at +50%, +100%).
- Capital Recycling (Free Roll).

Autopilot AI Logic

- Standard Plan: Sell 1/2 at Target 1, Move Stop to Breakeven.
- Trailing Stop: Tighten stop as price advances (Chandelier Exit).
- Time-Based Exit: If trade goes nowhere for 5 days, kill it.
- Panic Exit: Liquidity dry-up detection.

Operational Playbook

- Visualize "Planned Exits" on the chart involved.
- Alert when "Runner" position is active.
- Track "Points Left on Table" (Exited too early?).
- Automate "Breakeven" stop adjustment.

Risk & Contingency

- Risk: Whipsaw. Mitigation: Validated Support/Resistance checks.
- Risk: Gap over Stop. Mitigation: Defined Risk Options Spreads.
- Risk: Partial Fill on Exit. Mitigation: Market sweep logic.

Day-by-Day Schedule

Mon	Scale-Out Logic implementation.
Tue	ATR Trailing Stop engine.
Wed	Time-Decay Exit (Theta burn check).
Thu	Integration with Order Router.
Fri	Simulation of improved Exit Mechanics.

n8n Workflow Orchestration

Flows 5+	NoCode True	Sync Auto	Error 0
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Technical Architecture

- n8n container (Self-hosted).
- Webhooks: TradingView -> n8n -> API.
- Email Parser (IMAP -> JSON).
- Calendar Sync (Google Cal -> Market Schedule).

Autopilot AI Logic

- AI can trigger n8n workflows (e.g., "Research this symbol").
- n8n scrapes web/news and feeds context to AI.
- Notifications routing via n8n (Slack, Telegram, Email).
- Backup Scheduler (if internal scheduler dies).

Operational Playbook

- Secure n8n with Auth.
- Version control workflows (JSON export).
- Monitor webhook latency.
- Use n8n for "Human Approval" loops (Email with Yes/No link).

Risk & Contingency

- Risk: n8n crash. Mitigation: Restart policy always.
- Risk: Infinity loop. Mitigation: Execution limits.
- Risk: Security. Mitigation: Firewall internal API.

Day-by-Day Schedule

Mon	n8n Docker setup.
Tue	TradingView Webhook Handler.
Wed	News Scraping Workflow.
Thu	Daily Digest Email Workflow.
Fri	"Human in the Loop" approval flow.

Market Data Feeds v2 (Polygon.io/ThetaData)

Feed Polygon	Latency Low	Granularity Tick	Chain Fast
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Technical Architecture

- Polygon.io WebSocket Client (aiohttp).
- ThetaData adapter (for Options history).
- Feed Switcher (Failover logic).
- Aggregated bar builder (if using tick data).

Autopilot AI Logic

- AI needs "Order Book" (L2) visibility if possible.
- ThetaData provides superior Greeks history.
- Detect "Unusual Whales" activity (Block trades).
- Sub-second latency checks.

Operational Playbook

- Subscribe to Real-Time Options feed.
- Manage bandwidth (WebSocket compression).
- Compare vendor data quality.
- Cache massive historical requests.

Risk & Contingency

- Risk: Cost explosion. Mitigation: Cache aggressively.
- Risk: API Change. Mitigation: Adapter pattern.
- Risk: Bandwidth saturation. Mitigation: Filter symbols.

Day-by-Day Schedule

Mon	Polygon/ThetaData API integration.
Tue	WebSocket Stream Handler v2.
Wed	Historical Options Chain loader.
Thu	Unusual Options Activity scanner.
Fri	Latency comparison (Alpaca vs Polygon).

Machine Learning Feature Engineering

Features 50+	Pipeline Auto	Store Feature	Clean Yes
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Technical Architecture

- Feature Store (Parquet/Postgres).
- TA-Lib integration (RSI, MACD, Bollinger).
- Custom Features (Volatility Ratios, Skew).
- Normalization Pipeline (Z-Score, MinMax).

Autopilot AI Logic

- AI consumes "Feature Vector" not just raw bars.
- Features: "Distance from VWAP", "IV Percentile", "RSI Divergence".
- Lagged features (t-1, t-5) for prediction.
- Stationarity checks (ADF test) for ML models.

Operational Playbook

- Compute features EOD for training set.
- Compute features Real-Time for inference.
- Visualize Feature Importance (Random Forest).
- Detect Feature Drift.

Risk & Contingency

- Risk: Look-ahead in features. Mitigation: Shift ops correctly.
- Risk: Overfitting. Mitigation: Feature selection (Lasso).
- Risk: Compute lag. Mitigation: Vectorized pandas/numpy.

Day-by-Day Schedule

Mon	TA-Lib Wrapper & Common Indicators.
Tue	Custom Volatility Features.
Wed	Feature Store Schema.
Thu	Real-time calculation pipeline.
Fri	Feature Importance Analysis.

P&L; Attribution Dashboard

Alpha Calc	Beta Calc	Luck Isolated	View Clear
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Technical Architecture

- Attribution Engine (Brinson analysis style).
- Market Benchmark comparison (SPY returns).
- Tag-based P&L; (Long vs Short, Day vs Swing).
- Sharpe/Sortino per Strategy.

Autopilot AI Logic

- AI analyzes its own "Win Rate by Day of Week".
- Identify "Leaking Alpha" (e.g., stops too tight).
- Suggest "Stop Trading" for underperforming strategies.
- Compare Actual vs Expected Returns (Slippage impact).

Operational Playbook

- Weekly "Performance Review" report.
- Visualize "Equity Curve" vs "Benchmark".
- Drill down into "Losers" to find patterns.
- Track "Commissions Paid" as % of P&L;

Risk & Contingency

- Risk: Ego. Mitigation: Brutally honest data.
- Risk: Benchmark drift. Mitigation: Use correct benchmark (SPY vs QQQ).
- Risk: Data error. Mitigation: Reconcile with Broker.

Day-by-Day Schedule

Mon	Attribution Math logic.
Tue	Benchmark Data ingestion.
Wed	Tag-based Analysis (Strategy/Symbol).
Thu	Dashboard Widget implementation.
Fri	"Deep Dive" Report generation.

Strategy Optimization (Genetic Algos)

Gen 100	Pop 1000	Fit Improved	Overfit Checked
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Technical Architecture

- DEAP (Python Genetic Algo library).
- Fitness Function ($\text{Sharpe} * \text{WinRate}$).
- Population Manager (Crossover/Mutation).
- Distributed Worker Pool (Ray or Celery).

Autopilot AI Logic

- AI evolves strategies over the weekend.
- Genome: [EntryRSI, ExitProfit%, StopLossATR].
- Survival of the fused: Best params move to "Forward Test".
- Avoid "Local Minima" with high mutation rates initially.

Operational Playbook

- Run massive optimization jobs on cloud spot instances.
- Analyze "Parameter Plateaus" (Robustness).
- Don't auto-deploy: Require human review of "Evolved" params.
- Log Evolution History.

Risk & Contingency

- Risk: Curve Fitting (Extreme). Mitigation: OOS Validation mandatory.
- Risk: Compute Cost. Mitigation: Limit generations.
- Risk: Nonsense params. Mitigation: Strict bounds constraints.

Day-by-Day Schedule

Mon	Genetic Algo Engine (DEAP) setup.
Tue	Genome & Fitness Function definition.
Wed	Distributed Worker implementation.
Thu	Visualization of Evolution.
Fri	Run Evolution on Strategy #1.

Multi-Account Support

Accts >1	Alloc Split	Sync Perfect	View Uni
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Technical Architecture

- Account Manager Service (Auth mapping).
- Allocation Logic (Pro-rata vs Fixed).
- Order Splitter (Master Order -> Child Orders).
- Unified Dashboard View.

Autopilot AI Logic

- AI manages "IRA" (Conservative) vs "Margin" (Aggressive).
- Respect Wash Sale rules across accounts (Complex!).
- Different Risk Profiles per account.
- Aggregated Exposure check.

Operational Playbook

- Link multiple Broker API keys.
- Test "Broadcast" execution (1 signal -> 3 accounts).
- Verify "Partitioning" of data (Privacy).
- Handle "Partial Fills" across accounts (Fair allocation).

Risk & Contingency

- Risk: Cross-contamination. Mitigation: Strict DB isolation.
- Risk: Execution lag. Mitigation: Async parallel output.
- Risk: Wash Sales. Mitigation: Warning system.

Day-by-Day Schedule

Mon	Database Schema update (AccountID FK).
Tue	Broker Client Multiplexer.
Wed	Order Allocation/Splitter Logic.
Thu	Dashboard Multi-Account Selector.
Fri	Integration Test with 2 Mock Accounts.

Charting Enhancements (Annotations)

Markers Buy/Sell	Lines Support	Text Reason	Perf High
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Technical Architecture

- TradingView Lightweight Charts API extensibility.
- Annotation Layer (Markers, Lines, Areas).
- Backend "Chart Object" storage.
- Interactive Tooltips.

Autopilot AI Logic

- AI draws "Support Zones" it sees.
- Mark Entry/Exit prints on the chart automatically.
- Overlay "Fair Value" bands generated by model.
- Show "Pending Orders" as dashed lines.

Operational Playbook

- Sync annotations with backend state.
- Allow manual drawing/overrides.
- Persist chart state per symbol.
- Optimize rendering for thousands of candles.

Risk & Contingency

- Risk: UI Clutter. Mitigation: Toggles for layers.
- Risk: Browser crash. Mitigation: Canvas rendering.
- Risk: Mobile unreadable. Mitigation: Simplification mode.

Day-by-Day Schedule

Mon	Trade Markers (Arrows).
Tue	Support/Resistance Line rendering.
Wed	AI Interpretation Overlay (Text).
Thu	Interactive Order Modification (Drag & Drop).
Fri	Performance tuning.

Compliance Logging (Audit Trail)

Log Immutable	Search Fast	Detail Full	Export Easy
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Technical Architecture

- Immutable Audit Log (Append-only DB table).
- Change Data Capture (CDC) or Trigger-based logging.
- Full Snapshot storage of Decision Context.
- Admin Viewer.

Autopilot AI Logic

- Every decision must be reconstructable.
- Store: Price, Indicators, Prompt, Response, Time, Version.
- "Why did it buy?" must be answerable 1 year later.
- Hash chain for tamper-evidence (optional).

Operational Playbook

- Log rotation/archival to S3 (Glacier).
- GDPR/Privacy compliance (if handling user data).
- Searchable UI for "Incident Investigation".
- Export to JSON/CSV for regulators (or tax man).

Risk & Contingency

- Risk: Disk bloat. Mitigation: Compression.
- Risk: DB Performance. Mitigation: Separate Log DB.
- Risk: Missing logs. Mitigation: Async queue with retry.

Day-by-Day Schedule

Mon	Audit Schema Definition.
Tue	Middleware for Request/Response logging.
Wed	Autopilot Context Snapshot logic.
Thu	Admin Log Viewer UI.
Fri	Alter-check mechanics.

Live Paper Trading (Month 1 Review)

Trades >20	WinRate Analyzed	Drawdown Checked	Bugs Fixed
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Technical Architecture

- Performance Analysis Suite.
- Bug Triage Dashboard.
- Optimized Config based on feedback.
- Deployment Pipeline hardening.

Autopilot AI Logic

- Review AI behavior in live market conditions.
- Did slippage match backtests?
- Did the AI hesitate?
- Were there "Unknown Unknowns"?

Operational Playbook

- Daily "Standup" with the bot (Review logs).
- Tweak parameters live (Risk limits).
- Fix operational friction (UI annoyances).
- Prepare for Real Money (Capital Allocation).

Risk & Contingency

- Risk: Simulation artifacts. Mitigation: Treat paper as live.
- Risk: Overconfidence. Mitigation: Wait for stat sig sample.
- Risk: Burnout. Mitigation: Automation checks.

Day-by-Day Schedule

Mon	Month 1 Performance Aggregation.
Tue	Trade-by-Trade Post-Mortem.
Wed	System Stability Review.
Thu	Configuration Retuning.
Fri	"Go/No-Go" decision for Real Money.

Quarterly Review & Roadmap Adjustment

Plan Updated	TechDebt Paid	Team Ready	Goals Set
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Technical Architecture

- Refactoring Sprint.
- Documentation Update.
- Deprecation of unused features.
- Infrastructure upgrade planning.

Autopilot AI Logic

- Retrain models with collected data?
- New Strategy ideas based on Q3 observation?
- Enhance "Self-Reflection" capabilities.
- Upgrade LLM versions (new models released?).

Operational Playbook

- Clean up JIRA/Tasks.
- Archive Q3 logs/data.
- Update Master Plan (this document).
- Celebrate wins.

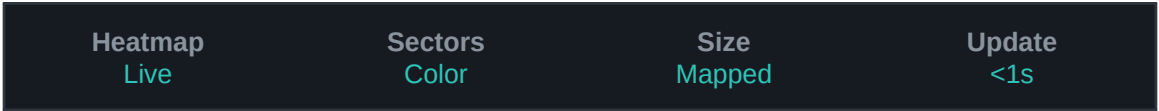
Risk & Contingency

- Risk: Stagnation. Mitigation: Set ambitious Q4 goals.
- Risk: Bloat. Mitigation: Remove dead code.
- Risk: Drift. Mitigation: Re-align with Vision.

Day-by-Day Schedule

Mon	Technical Debt Assessment.
Tue	Roadmap Review Session.
Wed	Architecture Review (Scalability).
Thu	Documentation Sprint.
Fri	Q4 Planning Finalization.

Portfolio Heat Maps & Advanced Visualization



Technical Architecture

- Treemap Visualization (d3-hierarchy or Recharts).
- Sector Aggregation Service.
- Color Scale Logic (Red/Green based on 1-day change).
- Size Logic (Position Value or Delta Exposure).

Autopilot AI Logic

- AI analyzes the Heatmap: "Tech is bleeding, Energy is green."
- Detect Sector Rotation early.
- Rebalance suggestions based on "Overheated" sectors.
- Visual confirmation of diversification.

Operational Playbook

- Dashboard Widget: "Market Map" style.
- Drill-down: Click Sector -> See Stocks.
- Filter by: P&L;, Delta, Gamma, Theta.
- Overlay Indices (SPY, QQQ) for relative strength.

Risk & Contingency

- Risk: Visual noise. Mitigation: Clean design.
- Risk: Lag. Mitigation: Aggregated backend endpoint.
- Risk: Colorblindness. Mitigation: Accessible palettes.

Day-by-Day Schedule

Mon	Aggregation Logic (Sector/Industry).
Tue	Treemap Component implementation.
Wed	Real-time update via WebSocket.
Thu	Interactive filtering.
Fri	Mobile optimization.

Advanced Risk Metrics (Sortino, Calmar, Omega)

Sortino >2	Calmar >1.5	Omega Calc	DownDev Track
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Technical Architecture

- Financial Math Library extension (empyrial or custom).
- Downside Deviation calculator (ignores upside vol).
- Tail Risk metrics (VaR, CVaR).
- Rolling Analysis windows.

Operational Playbook

- Add "Advanced Stats" tab to Dashboard.
- Report these metrics weekly.
- Compare against SPY/QQQ benchmarks.
- Explain metrics in tooltips.

Autopilot AI Logic

- Optimizing for Sortino (punish losses) vs Sharpe (punish all vol).
- If Calmar < 0.5, Halt Strategy (Drawdown too deep for return).
- Use Omega Ratio to assess probability of meeting threshold.
- Detect "Skewness" in return distribution.

Risk & Contingency

- Risk: Short history. Mitigation: Need 30+ data points.
- Risk: Calculation error. Mitigation: Unit test vs known library.
- Risk: Obsession. Mitigation: Returns pay bills, ratios don't.

Day-by-Day Schedule

Mon	Sortino Ratio implementation.
Tue	Calmar & Max Drawdown Logic.
Wed	Omega Ratio & Probability cones.
Thu	Rolling Metric logic.
Fri	Integration into Strategy Scoring.

SEC Filings Analysis (EDGAR)

Docs Parsed	Sentiment Extracted	Alpha Found	Delay <1m
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Technical Architecture

- SEC EDGAR API Wrapper (sec-edgar-downloader).
- XBRL Parser (for financial tables).
- LLM Summarizer (Long Context Window).
- Insider Trading Feed (Form 4).

Autopilot AI Logic

- AI reads "Risk Factors" section of 10-K.
- Detects "Material Weakness" disclosures.
- Scans Form 4 for "CEO Buying".
- Correlates Insider Buys with technical setup.

Operational Playbook

- Poll EDGAR RSS feed.
- Filter for S&P; 500 only initially.
- Store synthesized "Fundamental Score".
- Alert on "Late Filing" (NT 10-K) - huge red flag.

Risk & Contingency

- Risk: XBRL complexity. Mitigation: Use specialized lib.
- Risk: LLM Context limit. Mitigation: Chunking strategy.
- Risk: False positive. Mitigation: Human review loop.

Day-by-Day Schedule

Mon	EDGAR API Client.
Tue	Form 4 (Insider) Parser.
Wed	10-K/10-Q Text Extraction.
Thu	LLM Summarization Pipeline.
Fri	Integration with Strategy Weights.

Options Term Structure Analysis (3D Viz)

3D
Rendered

Skew
Live

Term
Tracked

Oppty
Alert

Technical Architecture

- 3D Surface Plotting (Plotly/Vis.js).
- Term Structure Calculator (IV vs Time).
- Constant Maturity IV Generator.
- Calendar Spread Scanner.

Operational Playbook

- Snapshot surface every 15 min.
- Rotate 3D chart in Dashboard.
- Highlight "Cheap" months to buy volatility.
- Overlay "Historical Average" surface.

Autopilot AI Logic

- Steep Contango? Sell front month, Buy back month (Calendars).
- Backwardation? Sell naked calls (carefully) or Put Spreads.
- Detect "Event Volatility" (Earnings) as a hump in the term structure.
- Normalize surface for historical comparison.

Risk & Contingency

- Risk: Bad data interpolation. Mitigation: Smooth splines.
- Risk: Illiquidity at far dates. Mitigation: Filter volume > 0.
- Risk: Browser crash (WebGL). Mitigation: Limit polygons.

Day-by-Day Schedule

Mon	Term Structure Data Pipeline.
Tue	3D Visualization Component.
Wed	Contango/Backwardation Logic.
Thu	Calendar Spread Opportunity Scanner.
Fri	Surface History Storage.

System Monitoring & Grafana Dashboards

Dash Live	Alerts Set	Log Graph	Uptime 99.9%
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Technical Architecture

- Prometheus (Metrics scraping).
- Grafana (Visualization).
- Loki (Log aggregation).
- FastAPI Middleware for Request Timing.

Autopilot AI Logic

- Auto-scale? (Not yet, but monitor load).
- Detect "Slow API" -> switch provider?
- Correlate "System Load" with "Market Volatility".
- Watchdog: If "Heartbeat" missing, restart container.

Operational Playbook

- Dashboard 1: Business Metrics (P&L, Trades).
- Dashboard 2: System Metrics (CPU, Memory, API Latency).
- Alert on 5xx Error Spikes.
- Monitor Disk Usage (Logs/DB).

Risk & Contingency

- Risk: Monitoring failure. Mitigation: External ping.
- Risk: Log explosion. Mitigation: Retention policy.
- Risk: Noise. Mitigation: Tuned thresholds.

Day-by-Day Schedule

Mon	Prometheus/Grafana Stack setup.
Tue	App Instrumentation (prometheus-client).
Wed	Log Aggregation (Loki/Promptail).
Thu	Dashboard Design.
Fri	Alert Rules configuration.

Multi-Strategy Portfolio Allocation

Alloc Dynamic	Div High	Cash Managed	Risk Unified
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Technical Architecture

- Capital Allocator Service.
- Mean-Variance Optimizer (Markowitz - Lite).
- Strategy Performance Tracker.
- Cash Management Logic.

Autopilot AI Logic

- Strategy A winning? Give it more capital?
- Strategy B losing? Cut allocation by 50%.
- Correlation check: Don't double down on Short Vol.
- Maintain Cash Buffer (20% reserve).

Operational Playbook

- Weekly Allocation Review.
- Rebalance logic (Threshold vs Time).
- Visualize "Allocation Pie Chart".
- Handle Margin Requirements dynamically.

Risk & Contingency

- Risk: Chasing performance. Mitigation: Lookback windows.
- Risk: Over-leverage. Mitigation: Portfolio Heat check.
- Risk: Strategy obsolescence. Mitigation: Sunset policy.

Day-by-Day Schedule

Mon	Capital Allocation Engine.
Tue	Performance-based rebalancing logic.
Wed	Mean-Variance Optimization prototype.
Thu	Cash Buffer Management.
Fri	Integration Test (Alloc changes).

AI-Powered Daily Briefing (Voice/Audio)

Audio Generated	TTS Natural	Content Relevant	Time 8am
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Technical Architecture

- TTS Engine (ElevenLabs/OpenAI Audio).
- Briefing Generator (LLM Summary).
- Audio File delivery (Telegram/Web).
- Podcast RSS Feed (Private).

Autopilot AI Logic

- Summarize Overnight Action (Futures/Asia/Europe).
- Highlight Key Risks/Events for today.
- Report Yesterday's P&L; and Trades.
- End with a "Market Sentiment" quote.

Operational Playbook

- Generate MP3 at 7:55 AM.
- Delivered via Mobile App/Telegram.
- Keep it under 2 minutes.
- Include "Wake Word" integration? (Optional).

Risk & Contingency

- Risk: Cost (TTS). Mitigation: Cache common phrases?
- Risk: Latency. Mitigation: Generate async.
- Risk: Annoyance. Mitigation: Opt-in.

Day-by-Day Schedule

Mon	Daily Briefing Text Template.
Tue	OpenAI TTS Integration.
Wed	Audio Stitching (Intro/Outro music).
Thu	Delivery channel (Telegram/S3).
Fri	"Persona" tuning (Jarvis vs Bloomberg).

Global Caching & Optimization

HitRate >90%	Lat <50ms	Store Redis	Inv Smart
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Technical Architecture

- Global Redis Cache Decorator.
- CDN for Frontend Assets (Cloudflare).
- Database Query Caching.
- Memoization of heavy math functions.

Autopilot AI Logic

- Don't re-calculate IV if price hasn't moved.
- Cache LLM responses for identical prompts.
- Prefetch data for likely engaged symbols.
- Invalidate cache on Event (WebSocket push).

Operational Playbook

- Monitor Cache Hit Ratio.
- Tune TTL (Time To Live) per data type.
- Ensure "Stale While Revalidate" pattern.
- Flush cache mechanisms.

Risk & Contingency

- Risk: Stale Data Trade. Mitigation: Market Data never cached > 1s.
- Risk: Memory leak. Mitigation: Redis MaxMemory policy.
- Risk: Complexity. Mitigation: Decorators.

Day-by-Day Schedule

Mon	Cache Strategy Audit.
Tue	Redis Pipeline Optimization.
Wed	Application-Level Caching (Memoization).
Thu	Browser Caching Headers.
Fri	Load Testing (Verify speedup).

Automated Strategy Research (Idea Generation)

Ideas Generated	Backtest Auto	Quality Ranked	Accept Rate
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Technical Architecture

- Research Agent (LLM + Stats).
- Idea Database.
- Auto-Backtester Bridge.
- Proposal Dashboard.

Autopilot AI Logic

- Read financial papers (Abstracts).
- Look for anomalies in historical data.
- Propose: "Buy SPY if RSI < 30 and VIX > 20".
- Run preliminary backtest and present results.

Operational Playbook

- Weekly "Research Report".
- User approves Promising Ideas for Deep Dive.
- Feedback loop: "Why did I reject this?"
- Crowdsource ideas from Twitter/Reddit sentiment?

Risk & Contingency

- Risk: Data Mining Bias. Mitigation: Strict holdout sets.
- Risk: Hallucinated Alpha. Mitigation: Math check.
- Risk: Spam. Mitigation: Quality filters.

Day-by-Day Schedule

Mon	Research Agent Prompting.
Tue	Hypothesis Generation Logic.
Wed	Auto-Backtest Binding.
Thu	Opportunity Ranking System.
Fri	"Shark Tank" UI (Approve/Reject).

Advanced Position Sizing (Kelly Criterion)

Kelly F	Size Dynamic	Risk Capped	Growth Geo
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Technical Architecture

- Kelly Criterion Calculator (Continuous).
- Half-Kelly implementation (Safety).
- Volatility Scaled Sizing.
- Correlation-Adjusted sizing.

Autopilot AI Logic

- Win Rate * Win/Loss Ratio input.
- Adjust size based on "Conviction Score".
- Never exceed generic Max Limit (e.g., 5%).
- Scale down in drawdowns (Anti-Martingale).

Operational Playbook

- Review Sizing logic impact on backtests.
- Visualize "Bet Size" vs "Edge".
- Override limits for manual trades.
- Explain "Why" size was chosen in logs.

Risk & Contingency

- Risk: Full Kelly = Ruin volatility. Mitigation: Use Fractional Kelly (0.3 - 0.5).
- Risk: Estimation error. Mitigation: Conservative inputs.
- Risk: Fat tails. Mitigation: Hard caps.

Day-by-Day Schedule

Mon	Kelly Formula Implementation.
Tue	Volatility Scaling Logic.
Wed	Correlation Adjustment.
Thu	Integration with Order Router.
Fri	Backtest Comparison (Fixed vs Kelly).

Security Hardening & Penetration Test

Vuln 0	Auth MFA	Keys Rotated	Audit Pass
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Technical Architecture

- Vault (Hashicorp) or AWS Secrets Manager.
- IP Whitelisting.
- Rate Limiting (Stricter).
- Dependency Audit (Snyk/Dependabot).

Autopilot AI Logic

- Detect abnormal access patterns.
- Alert on "New IP" login.
- Sanitize all inputs (SQL Injection prevention).
- Verify TLS certificate validity.

Operational Playbook

- Rotate all API keys.
- Enable MFA for Admin Dashboard.
- Run OWASP ZAP scan against localhost.
- Review Access Logs.

Risk & Contingency

- Risk: Locked out. Mitigation: Recovery codes.
- Risk: Key leak. Mitigation: Revocation plan.
- Risk: Convenience vs Security. Mitigation: User friction balance.

Day-by-Day Schedule

Mon	Secrets Management Migration.
Tue	Network Security (Firewall/IP).
Wed	Dependency Vulnerability Scan.
Thu	Penetration Testing (Self).
Fri	Incident Response Plan drill.

Documentation Overhaul (Sphinx/MkDocs)

Docs 100%	Search Yes	API Spec	Guides Video
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Technical Architecture

- Sphinx/MkDocs site generator.
- Docstrings (Google Style).
- Auto-generated API Docs (Swagger/Redoc).
- Architecture Diagrams (Mermaid.js).

Autopilot AI Logic

- Document the "Decision Tree" logic.
- Explain "Prompt Strategy" in wiki.
- Maintain "Known Issues" list.
- Video walkthroughs of dashboard.

Operational Playbook

- Build docs on CI dispatch.
- Host on internal server (or GitHub Pages private).
- Write "New Developer" onboarding guide.
- Document Disaster Recovery proc.

Risk & Contingency

- Risk: Stale docs. Mitigation: CI check for doc coverage.
- Risk: Secret leakage in docs. Mitigation: Scan.
- Risk: Time sink. Mitigation: Focus on "Why" not "What".

Day-by-Day Schedule

Mon	Docstring Audit.
Tue	Architecture Diagrams Update.
Wed	Operational Runbooks.
Thu	API Documentation Polish.
Fri	Site Launch & Review.

Year 1 Retrospective & Calibration

Y1 P&L; Final	Lessons Logged	Roadmap Y2	Rest Yes
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Technical Architecture

- Year-End Report Generator.
- Tax Lot Optimizer.
- Archive Service (Year 1 Data).
- Calibration Engine.

Autopilot AI Logic

- Re-train models on full Year 1 dataset.
- Identify "Seasonality" in Year 1 performance.
- Adjust "Risk Appetite" for Year 2.
- Celebrate automation success.

Operational Playbook

- Generate Tax Documents (Preview).
- Archive Logs to Cold Storage.
- Plan Hardware Upgrades?
- Team Party (You and the Bot).

Risk & Contingency

- Risk: Burnout. Mitigation: Take a week off.
- Risk: Market shift. Mitigation: Continuous learning.
- Risk: Complacency. Mitigation: Verify backups.

Day-by-Day Schedule

Mon	Year-End Performance Analysis.
Tue	Tax Optimization Review.
Wed	Archival & Cleanup.
Thu	Year 2 Roadmap Finalization.
Fri	NEW YEAR'S EVE - System Shift.

Multi-Agent System Architecture (Swarm)

Agents 3+	Chat Internal	Consensus Vote	Speed Parallel
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Technical Architecture

- Agent Orchestrator (LangGraph / CrewAI pattern).
- Shared Memory (Vector DB + Redis).
- Inter-Agent Messaging Protocol.
- Supervisor Agent (Meta-Cognition).

Autopilot AI Logic

- Role 1: Researcher (Reads news, finds setup).
- Role 2: Risk Manager (Validates setup).
- Role 3: Executor (Optimizes entry).
- Supervisor: Resolves disputes between Research and Risk.

Operational Playbook

- Log internal "Chat" between agents for debugging.
- Visualize Agent State (Thinking... Waiting...).
- Parallelize Agent execution (Async).
- Handle "Deadlock" where agents disagree forever.

Risk & Contingency

- Risk: Complexity explosion. Mitigation: Strict interfaces.
- Risk: Cost (many LLM calls). Mitigation: Smaller models for sub-agents.
- Risk: Latency. Mitigation: AsyncIO.

Day-by-Day Schedule

Mon	Agent Base Class & Message Bus.
Tue	Research Agent implementation.
Wed	Risk Agent implementation.
Thu	Supervisor Logic (Voting).
Fri	Integration Test: The Debate.

Deep Research Agent (Web & Paper Reading)

Sources 20+	Summary Clean	Alpha Cited	Auto Daily
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Technical Architecture

- Browser Tool (Playwright/Selenium headless).
- PDF Parser (Unstructured.io).
- Recursive Summarizer (Map-Reduce).
- Fact-Checker Module.

Autopilot AI Logic

- Scrape financial blogs, Substack, Reddit.
- Read Whitepapers (Crypto) or Earnings Call Transcripts.
- Cross-reference claims ("Is this rumor true?").
- Output: "Confidence Score" on information.

Operational Playbook

- Schedule "Deep Dives" overnight.
- Output a "Morning Brief" for the Human.
- Store Knowledge Graph of concepts.
- Rate limit web scraping strictly.

Risk & Contingency

- Risk: Bot detection. Mitigation: Rotation proxies.
- Risk: Fake News. Mitigation: Source reputation list.
- Risk: IP Ban. Mitigation: Respect robots.txt.

Day-by-Day Schedule

Mon	Scraper Infrastructure.
Tue	PDF/Text Extraction pipeline.
Wed	Summarization Logic (LLM).
Thu	Knowledge Graph Builder.
Fri	"Analyst Report" Generation.

The "Devil's Advocate" Risk Agent

Killed Tracking	Saved \$\$\$	Audit Why	Bias Checked
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Technical Architecture

- Contrarian Prompt Engineering.
- Stress Test Scenario Runner.
- Bias Detector (Confirmation Bias).
- Veto Power Module.

Autopilot AI Logic

- Strategy prompts: "Buy AAPL".
- Risk Agent prompts: "Tell me 5 reasons why AAPL will crash".
- Agent checks for "Overcrowded Trade".
- Agent checks for "Macro Headwinds".

Operational Playbook

- Every trade MUST pass the Devil's Advocate.
- Log the "Argument" against the trade.
- Human can override, but must type "I override the risk agent".
- Track "Saved Capital" (Trades killed that would have lost).

Risk & Contingency

- Risk: Paralysis (Too strict). Mitigation: Confidence threshold.
- Risk: Hallucination. Mitigation: Grounding.
- Risk: Latency. Mitigation: Timeout.

Day-by-Day Schedule

Mon	Contrarian Persona definition.
Tue	Macro Data integration for Risk Agent.
Wed	Scenario Injection.
Thu	Veto Logic implementation.
Fri	Backtest: With vs Without Risk Agent.

Execution Agent (Snippet Optimization)

Slippage
Neg

Improve
bps

Dark
Pool

Hide
Size

Technical Architecture

- Micro-structure Analyzer (L2 Data).
- Order Type Selector (Iceberg, PEG, IOC).
- Liquidity Seeker.
- Latency Arbitrage protection.

Autopilot AI Logic

- Don't just show size. Hide in the noise.
- Detect "Algo Predation" (HFT sniffing).
- Route to different exchanges (if Crypto/Futures).
- Time execution for "Low Vol" microseconds.

Operational Playbook

- Measure "Implementation Shortfall".
- A/B test Execution Algorithms.
- Monitor "Rejection Rate".
- Visualize "Fill Quality" heatmaps.

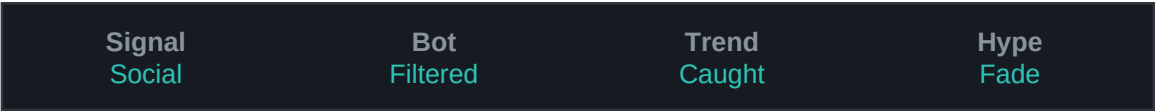
Risk & Contingency

- Risk: Over-engineering. Mitigation: Keep it simple for small size.
- Risk: Exchange fees. Mitigation: Fee-aware routing.
- Risk: Stuck orders. Mitigation: Aggressive cancel.

Day-by-Day Schedule

Mon	Order Book (L2) Analysis logic.
Tue	Iceberg Order implementation.
Wed	"Predator Detection" logic.
Thu	Smart Router logic.
Fri	Execution Quality Report.

Sentiment Agent (Social Media/News)



Technical Architecture

- Twitter/X API Client (or Scraper).
- Reddit API (PRAW).
- Discord Scraper (Self-bot).
- NLP Sentiment Engine (FinBERT).

Autopilot AI Logic

- Detect "Trending Tickers" before price moves.
- Filter out "Bot Spam" (\$CASHTAG spam).
- Analyze "Retail Sentiment" vs "Institutional Flow".
- Fade key retail hype? Or ride it?

Operational Playbook

- Monitor specific "Alpha" accounts.
- Track "Mentions Velocity" (rate of change).
- Visualize Word Clouds.
- Alert on "Viral" news.

Risk & Contingency

- Risk: API Ban. Mitigation: Official APIs or rotation.
- Risk: Noise. Mitigation: High filters.
- Risk: Pump & Dump. Mitigation: Volatility gates.

Day-by-Day Schedule

Mon	Social Data Ingestion.
Tue	Spam/Bot Filtering Logic.
Wed	FinBERT Integration.
Thu	"Hype Score" calculation.
Fri	Backtest: Social Signal vs Price.

Macro Agent (Fed/Economic Data)

GDP Tracked	Rates Tracked	Regime Inf	Asset Alloc
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Technical Architecture

- FRED (Federal Reserve) API.
- BLS (Labor Stats) Scraper.
- Treasury Yield Curve analyzer.
- Regime Classifier (Inflationary vs Deflationary).

Autopilot AI Logic

- If Yield Curve Inverted -> Reduce Beta.
- If Inflation Rising -> Value over Growth.
- Analyze Fed Speak (Hawk vs Dove).
- Adjust Portfolio Allocation based on Macro Regime.

Operational Playbook

- Update Macro Dashboard weekly.
- Alert on "Surprise" data prints (NFP).
- Correlate Assets to Macro factors.
- Scenario Planning (Recession simulation).

Risk & Contingency

- Risk: Lagging data. Mitigation: Market reaction is real-time.
- Risk: False signal. Mitigation: Multi-factor confirmation.
- Risk: Complexity. Mitigation: Simple heuristics.

Day-by-Day Schedule

Mon	FRED API integration.
Tue	Yield Curve logic.
Wed	Regime Classification model.
Thu	Allocation Adjustment logic.
Fri	Macro Dashboard.

Crypto Market Integration (24/7)

Exch Binance	Data Stream	Exec Live	Sleep No
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Technical Architecture

- CCXT Library (Unified Crypto API).
- Binance/Coinbase/Kraken Adapters.
- 24/7 Scheduler modifications.
- WebSocket Keep-alive logic.

Autopilot AI Logic

- Crypto moves fast. Latency is critical.
- Arbitrage opportunities exist.
- Detect "Whale Support" levels on-chain.
- Manage Funding Rates (Perp Futures).

Operational Playbook

- Separate "Crypto" portfolio bucket.
- Handle 8 decimal places (Satoshis).
- Monitor "Exchange Health" (Withdrawals paused?).
- Tax Lots for Crypto (FIFO/LIFO).

Risk & Contingency

- Risk: Exchange Hack. Mitigation: Self-custody or spread checks.
- Risk: Tether depeg. Mitigation: Stablecoin diversity.
- Risk: Volatility. Mitigation: Lower leverage.

Day-by-Day Schedule

Mon	CCXT Integration & Config.
Tue	Exchange Connectivity Tests.
Wed	Market Data Normalization (Crypto vs TradFi).
Thu	Funding Rate Scraper.
Fri	First Paper Trade (BTC/USDT).

Arbitrage Scanner (Cross-Exchange)

Arb Found	Spread >0.5%	Exec Atomic	Risk Transfer
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Technical Architecture

- Multi-Exchange Order Book Aggregator.
- Latency Normalization.
- Transfer Time Estimator.
- Execution Coordinator (Leg 1 & Leg 2).

Autopilot AI Logic

- Spot Price Discrepancy (Binance vs Kraken).
- Funding Rate Arb (Long Spot / Short Perp).
- Triangular Arb (BTC -> ETH -> USDT -> BTC).
- Execution risk is the killer.

Operational Playbook

- Monitor "Withdrawal Fees" and "Transfer Times".
- Balance rebalancing logic (inventory management).
- Visualize Spreads live.
- Auto-disable if volatility spike.

Risk & Contingency

- Risk: Leg 1 fills, Leg 2 fails. Mitigation: Market orders (eat spread).
- Risk: Withdrawal stuck. Mitigation: Keep inventory on both.
- Risk: Ban. Mitigation: Respect limits.

Day-by-Day Schedule

Mon	Aggregator Logic.
Tue	Fee Calculator (Taker/Maker + Withdrawal).
Wed	Inventory Rebalancer.
Thu	Execution Logic (Parallel).
Fri	Simulation on Historical Data.

DeFi Integration & Flash Loans

Chain ETH/SOL	Dex UniV3	Loan Flash	MEV Avoid
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Technical Architecture

- Web3.py / Ethers.js integration.
- Local Node (Geth/Reth) or Infura.
- Smart Contract Interaction (ABI).
- MEV Protection (Flashbots).

Autopilot AI Logic

- Monitor DEX Liquidity Pools.
- Calculate Impermanent Loss risk.
- Execute Flash Loan Arb (Atomic transaction).
- Gas Fee optimization.

Operational Playbook

- Wallet Management (Private Keys secure!).
- Gas Price Estimator (EIP-1559).
- Approve Token allowances.
- Monitor Contract Risk (Rugpull scanner).

Risk & Contingency

- Risk: Smart Contract Bug. Mitigation: Audited protocols only.
- Risk: Private Key theft. Mitigation: HSM / Hardware Wallet.
- Risk: Failed Tx cost. Mitigation: Gas limit.

Day-by-Day Schedule

Mon	Web3 Connection & Wallet Setup.
Tue	DEX Pricing Oracle.
Wed	Flash Loan Contract Template.
Thu	MEV/Flashbots integration.
Fri	Testnet Deployment.

Social Trading & Copy-Trading

Leader Follow	Slippage Copy	Scale Ratio	Profit Shared
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Technical Architecture

- Copy-Trading Engine.
- Leaderboard Scraper (eToro/Binance).
- Signal Normalizer.
- Ratio Sizer.

Autopilot AI Logic

- Identify "Consistently Profitable" traders.
- Filter out "Lucky Gamblers" (High variance).
- Replicate trades with proportional sizing.
- Fade "Inverse Jim Cramer" logic.

Operational Playbook

- Monitor "Leader" drift.
- Alert on "Strategy Change".
- Stop Loss on the Copy Portfolio.
- Report Copy Performance vs Benchmark.

Risk & Contingency

- Risk: Leader blows up. Mitigation: Hard stop per leader.
- Risk: Front-running. Mitigation: Random delay?
- Risk: Liquidity constraint. Mitigation: Cap size.

Day-by-Day Schedule

Mon	Leader Scraper.
Tue	Performance Analysis metrics.
Wed	Copy Engine logic.
Thu	Sizing & Ratio logic.
Fri	Simulation: Copy Top 10.

Leaderboard & Gamification

Rank Global	Badge Earned	Streak Days	Fun High
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Technical Architecture

- Gamification Service (XP, Levels).
- Achievement System (e.g., "Diamond Hands").
- Public (or Private) Leaderboard UI.
- Weekly Challenge generator.

Autopilot AI Logic

- Reward "Disciplined Trading" (Following plan).
- Penalize "FOMO" (Breaking rules).
- Generate "Badges" for milestones.
- Compare vs "Market Average".

Operational Playbook

- Display Level/XP on Dashboard.
- Unlock "Features" at higher levels?
- Celebrate "Win Streaks".
- Analyze behavioral impact.

Risk & Contingency

- Risk: Over-trading for XP. Mitigation: Reward ROI not Volume.
- Risk: Distraction. Mitigation: Keep it subtle.
- Risk: Gaming the system. Mitigation: Audited metrics.

Day-by-Day Schedule

Mon	Gamification Schema (XP/Level).
Tue	Achievement Definitions.
Wed	Leaderboard Backend.
Thu	UI Integration.
Fri	"Quest" System.

Mobile App v2 (Push Notifications)

App Installed	Push Instant	Chart Touch	Auth Bio
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Technical Architecture

- React Native (or PWA improvements).
- Push Notification Service (Firebase/Expo).
- Biometric Auth (FaceID).
- Mobile-Optimized Charts.

Autopilot AI Logic

- Send "Actionable Notifications" (Buy/Sell buttons).
- Summarize charts for small screens.
- Voice Dictation input ("Buy 100 AAPL").
- Location-aware security.

Operational Playbook

- Deploy to TestFlight / Play Store (Internal).
- Optimize battery usage.
- Ensure "Offline Mode" works (Read-only).
- Sync state perfectly with Desktop.

Risk & Contingency

- Risk: Phone theft. Mitigation: Bio-auth every open.
- Risk: Fat finger. Mitigation: Confirm dialogs.
- Risk: Network drop. Mitigation: Robust retry.

Day-by-Day Schedule

Mon	React Native init.
Tue	Biometric Auth.
Wed	Push Notification handler.
Thu	Chart Interaction (Touch).
Fri	Build & Deploy.

Conversational AI Interface (Voice)

STT Whisper	TTS Eleven	Lat <1s	Context Held
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Technical Architecture

- Whisper API (OpenAI) for STT.
- Intent Classifier (LLM).
- Action Executor.
- ElevenLabs TTS return.

Autopilot AI Logic

- User: "How is my tech exposure?" -> AI: "You are 40% tech..."
- User: "Close all loosing trades." -> AI: "Confirm close 3 positions?"
- Understand context ("What about AAPL?" refers to previous q).
- Voice Authentication check.

Operational Playbook

- Wake word detection (Porcupine).
- Microphone integration in Browser/Mobile.
- Visualizer (Waveform).
- Privacy mode (Mute).

Risk & Contingency

- Risk: Misinterpretation. Mitigation: Confirmation required.
- Risk: Latency. Mitigation: Stream text first.
- Risk: Noise. Mitigation: Noise cancellation.

Day-by-Day Schedule

Mon	Whisper Integration.
Tue	Intent Classification Pipeline.
Wed	Action Mapping.
Thu	TTS Response.
Fri	Voice UI Visualization.

HFT Infrastructure (Kernel Bypass/FPGA)

Lat <10us	Lang Rust	NIC Solar	Jitter 0
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Technical Architecture

- Rust/C++ Market Data Handler (Zero-copy).
- Kernel Bypass Networking (DPDK/Solarflare).
- Direct Market Access (DMA) protocols (FIX/SBE).
- Ring Buffer visualization.

Autopilot AI Logic

- AI designs the strategy, Rust executes it.
- Logic must be branchless if possible.
- Pre-calc lookup tables.
- Hardware timestamps synchronization.

Operational Playbook

- Co-location considerations (Equinix NY4).
- Isolate CPUs (taskset).
- Disable Interrupts on trading cores.
- Measure "Tick-to-Trade" latency strictly.

Risk & Contingency

- Risk: Runaway Algo. Mitigation: Hardware-level kill switch.
- Risk: Integer Overflow. Mitigation: Checked math.
- Risk: Cost. Mitigation: Only needed for specific arbs.

Day-by-Day Schedule

Mon	Rust environment setup.
Tue	TCP Neutralization (Solarflare mocking).
Wed	FIX Protocol Parser optimization.
Thu	Ring Buffer Queue implementation.
Fri	Latency Benchmarking (Round Trip).

Sentiment Arbitrage (News -> Trade)

Speed <50ms	NLP Fast	Alpha High	False Low
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Technical Architecture

- Low-Latency News Feed (Benzinga/Bloomberg).
- Keyword Spotter (Regex > LLM for speed).
- Event Classifier (Guidance Raise/Lower).
- Gap Fading Logic.

Autopilot AI Logic

- "Earnings Beat" + "Guidance Raise" = Instant Buy.
- "SEC Investigation" = Instant Sell.
- Fade the initial spike (Mean Reversion)?
- Filter out "Re-hashed" news.

Operational Playbook

- Milliseconds matter here.
- Pre-map Tickers to Keywords.
- Simulate "Squawk Box" audio processing?
- Visualize News Velocity.

Risk & Contingency

- Risk: Algo reading wrong tag. Mitigation: Confidence score.
- Risk: Liquidity gap. Mitigation: Limit orders with offset.
- Risk: Late entry. Mitigation: Cancel if > 200ms.

Day-by-Day Schedule

Mon	Fast News Feed API.
Tue	Regex Keyword Engine (Aho-Corasick).
Wed	Gap Logic.
Thu	Backtest: Headline vs 1-min Bar.
Fri	Live Paper Run.

Pairs Trading (Cointegration StatArb)

Pairs 50+	Coint Test	Z-Score Entry	Hedge Ratio
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Technical Architecture

- Cointegration Tester (Engle-Granger).
- Kalman Filter for dynamic Hedge Ratio.
- Spread Diff Calculator.
- Half-Life Analyzer.

Autopilot AI Logic

- Scan SP500 for Cointegrated Pairs (daily).
- If Z-Score > 2.0, Short Winner / Long Loser.
- Exit at Z-Score = 0 (Mean Reversion).
- Adjust for Dividends/Splits.

Operational Playbook

- Visualize the "Spread" chart.
- Maintain "Neutral" market exposure.
- Alert if Cointegration breaks (Regime shift).
- Manage "Legging In" risk.

Risk & Contingency

- Risk: Spread divergence (M&A;). Mitigation: News filter.
- Risk: Execution lag. Mitigation: Atomic orders.
- Risk: Hard to borrow. Mitigation: Inventory check.

Day-by-Day Schedule

Mon	Cointegration Math library.
Tue	Kalman Filter implementation.
Wed	Z-Score trigger logic.
Thu	Pair Selector Scanner.
Fri	Backtest: GOOG vs GOOGL.

Long/Short Equity (Factor Models)

Beta 0	Alpha High	Vol Low	Factors 5
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Technical Architecture

- Factor Database (Value, Momentum, Quality, Size, Vol).
- Risk Model (Barra-style).
- Optimizer (Maximize Factor tilt, Minimize Beta).
- Rebalance Engine.

Autopilot AI Logic

- Rank Universe by "Quality" and "Momentum".
- Long Top Decile, Short Bottom Decile.
- Constrain Sector Weights (Sector Neutral).
- Rebalance Monthly or Quarterly.

Operational Playbook

- Data intensive (Fundamentals required).
- Visualization: Factor Exposure Radar Chart.
- Monitor "Factor Crowding".
- Cost of Carry calculation (Shorts).

Risk & Contingency

- Risk: Factor crash (Momentum unwind). Mitigation: Factor timing?
- Risk: Short Squeeze. Mitigation: Stop loss on shorts.
- Risk: Data lag. Mitigation: Point-in-time data.

Day-by-Day Schedule

Mon	Factor Definition & Calculation.
Tue	Ranking Engine.
Wed	Optimizer (SciPy/CVXPY).
Thu	Sector Neutrality constraints.
Fri	Backtest: L/S Equity 2010-2025.

VIX Trading & Volatility of Volatility

VVIX Track	Contango Yield	Roll Cost	Hedge Tail
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Technical Architecture

- VIX Term Structure analyzer.
- VIX Futures Roll Yield calculator.
- VVIX (Vol of Vol) signal.
- ETN Decay model (VXX/UVXY).

Autopilot AI Logic

- If VIX Term Structure in Contango -> Short VIX (carry).
- If VVIX spikes -> Buy VIX Calls (Tail protection).
- Mean Reversion of VIX to 15-20.
- Hedge Equity Longs with VIX Calls.

Operational Playbook

- Handle VIX Futures expiration (Cash Settled).
- Monitor Roll Cost (Headwind).
- Alert on "VIX Inversion".
- Visualize Term Structure.

Risk & Contingency

- Risk: VIX explosion (Volmageddon). Mitigation: Defined risk spreads.
- Risk: ETN delisting. Mitigation: Trade Futures directly.
- Risk: Roll bleed. Mitigation: Timing.

Day-by-Day Schedule

Mon	VIX Futures Data Feed.
Tue	Roll Yield Logic.
Wed	VVIX Signal generation.
Thu	ETN Arb logic.
Fri	Simulation: Short VIX strategy.

Gamma Scalping (Dynamic Hedging)

Gamma Long	Theta Paid	Scalp Freq	PnL Smooth
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Technical Architecture

- Continuous Delta Hedger.
- Gamma Profile visualizer.
- Re-hedge Trigger (Time vs Price move).
- PnL Attribution (Gamma vs Theta).

Autopilot AI Logic

- Long Straddle (Long Gamma). Delta is neutral initially.
- Price moves up -> Delta becomes positive -> Sell Stock to neutral.
- Price moves down -> Delta becomes negative -> Buy Stock to neutral.
- Buy Low, Sell High automatically.

Operational Playbook

- Execution costs kill this strategy. Low fees mandatory.
- Balance "Hedge Frequency" vs "Transaction Cost".
- Trade liquid underlyings (SPY/QQQ).
- Monitor "Breakeven" volatility.

Risk & Contingency

- Risk: Low Volatility (Theta bleed). Mitigation: Short Term scalps.
- Risk: Gap Risk. Mitigation: Gamma creates profit on gaps.
- Risk: Execution Slippage. Mitigation: Limit orders.

Day-by-Day Schedule

Mon	Delta Hedging Engine.
Tue	Trigger Logic (Fixed Delta band).
Wed	PnL Decomposition.
Thu	Simulation with Transaction Costs.
Fri	Live Test (Small Size).

Reinforcement Learning (PPO Agent)

Reward Max	Episode 1M	Policy Stable	Chaos High
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Technical Architecture

- OpenAI Gym Env (Custom TradingEnv).
- Stable Baselines 3 (PPO/A2C).
- Reward Function Engineering.
- Observation Space Definition.

Autopilot AI Logic

- Observation: [Returns, RSI, DOM, Holdings].
- Action: [Buy, Sell, Hold, Size].
- Reward: Sharpe Ratio change.
- Train on 10 years of Tick Data.

Operational Playbook

- Training takes days on GPU.
- Visualize "Learning Curve".
- Check for "Gaming the Reward" (e.g., holding cash only).
- Sim-to-Real gap measurement.

Risk & Contingency

- Risk: Overfitting to noise. Mitigation: Regularization.
- Risk: Catastrophic forgetting. Mitigation: Replay buffer.
- Risk: Unexplainable actions. Mitigation: Feature attribution.

Day-by-Day Schedule

Mon	Gym Environment Setup.
Tue	Reward Function Design.
Wed	PPO Agent Implementation.
Thu	Hyperparameter Tuning.
Fri	Evaluation vs Benchmark.

Dark Pool Analysis & Hidden Liquidity

Dark Vol	Block Found	Level Marked	Fade Ready
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Technical Architecture

- Dark Pool Print Aggregator (FINRA TRF).
- Volume Profile (Lit vs Dark).
- Support/Resistance Zones based on Blocks.
- Interaction Algo.

Autopilot AI Logic

- Large Dark Print at Price X -> Likely Support/Resistance.
- Price approaches X -> Expect bounce.
- Distinguish "Sell" vs "Buy" prints (Tick rule).
- Filter out "Late Prints".

Operational Playbook

- Visualize "Ghost Bars" on chart.
- Alert on "Signature Prints" (unusual size).
- Correlate with Options Flow.
- Store Historical Dark Levels.

Risk & Contingency

- Risk: Delayed reporting (up to 15 min). Mitigation: Awareness.
- Risk: Misinterpretation. Mitigation: Context.
- Risk: Noise. Mitigation: Size filters.

Day-by-Day Schedule

Mon	FINRA TRF Data Feed.
Tue	Lit vs Dark volume splitter.
Wed	Key Level Identifier.
Thu	Interaction Strategy logic.
Fri	Visualization.

Event-Driven Strategies (Merger Arb/Spinoffs)

Deal Spread	Prob Calc	Risk Binary	Date Track
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Technical Architecture

- M&A; News Scraper.
- Deal Spread Calculator.
- Probability Implied Calculator.
- Legal Document Parser.

Operational Playbook

- Maintain "Deal List".
- Monitor "Spread Tightening/Widening".
- Close before "Judge Ruling" (Binary risk).
- Handle Cash/Stock mix deals.

Autopilot AI Logic

- Deal Announced: Company A buys B for \$50.
- Price B trades at \$48.
- AI calculates Annualized Return of the \$2 spread.
- Check Antitrust Risk (LLM reads news).

Risk & Contingency

- Risk: Deal Break. Mitigation: Position size (Diversify).
- Risk: Timeline extension. Mitigation: ROI re-calc.
- Risk: Bidding War. Mitigation: Upside optionality.

Day-by-Day Schedule

Mon	Deal Feed integration.
Tue	Spread Math & ROI calc.
Wed	Antitrust Sentiment Analysis.
Thu	Portfolio Construction (Merger Basket).
Fri	Deal Break Simulation.

Weather Derivatives & Commodities

Temp Obs	NatGas Corr	HDD Calc	Agri Check
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Technical Architecture

- NOAA Weather API.
- Commodity Futures Feed (NatGas, Corn).
- Correlation Engine (Weather -> Price).
- Seasonality Mapper.

Autopilot AI Logic

- Forecast: "Cold Snap in Northeast".
- Action: Buy Natural Gas Futures (Heating demand).
- Forecast: "Drought in Midwest".
- Action: Buy Corn/Soybeans.
- Check WASDE reports.

Operational Playbook

- Visualize Weather Maps overlay.
- Track "HDD/CDD" deviations from normal.
- Monitor Storage Reports (EIA).
- Manage Futures Roll.

Risk & Contingency

- Risk: Forecast wrong. Mitigation: Stop loss.
- Risk: Supply shock (War). Mitigation: News filter.
- Risk: Limit moves. Mitigation: Options definition.

Day-by-Day Schedule

Mon	NOAA API & GFS Model data.
Tue	HDD/CDD Metric calculation.
Wed	NatGas Correlation logic.
Thu	Visual Dashboard (Heatmaps).
Fri	Strategy: Weather vs Price.

Order Flow Toxicity & VPIN

VPIN High	Toxic Yes	Pull Quotes	Safe Mode
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Technical Architecture

- VPIN (Volume-Synchronized Probability of Informed Trading) calculator.
- Order Imbalance Monitor.
- Toxic Flow Detector.
- Liquidity Provider Protection.

Autopilot AI Logic

- If VPIN spikes -> Informed Traders are present.
- Action: Widen spreads or Stop Trading.
- Don't provide liquidity to toxic flow.
- Detect "Adverse Selection".

Operational Playbook

- High-speed calculation (Volume bucketing).
- Alert on "Flash Crash" conditions.
- Visualize Toxicity heat.
- Review limit order fills (are we getting run over?).

Risk & Contingency

- Risk: False positive. Mitigation: Threshold tuning.
- Risk: Lag. Mitigation: Real-time tick processing.
- Risk: Market maker obligation? (None for us).

Day-by-Day Schedule

Mon	VPIN Theory & Formula.
Tue	Volume Bucketing Engine.
Wed	Imbalance Logic.
Thu	Protection Triggers.
Fri	Toxicity limit stress test.

Market Making (Providing Liquidity)

Spread Captured	Rebate Earned	Inv Flat	Vol High
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Technical Architecture

- Market Making Bot (Quote on both sides).
- Inventory Manager (Skew quotes to flatten).
- Rebate Calculator.
- Micro-price model.

Autopilot AI Logic

- Quote Bid/Ask based on Micro-price + Spread.
- Inventory gets Long? Lower Bid/Ask to encourage selling.
- Inventory gets Short? Raise Bid/Ask to encourage buying.
- Target: Net zero position EOD.

Operational Playbook

- Require limit orders (Maker).
- Capture Exchange Rebates (if applicable).
- Monitor "Position Hold Time" (Seconds).
- Avoid trending markets (Delta risk).

Risk & Contingency

- Risk: Inventory accumulation. Mitigation: Aggressive skew.
- Risk: Adverse selection. Mitigation: VPIN filter (Wk 76).
- Risk: Tech failure. Mitigation: Cancel on Disconnect.

Day-by-Day Schedule

Mon	MM Logic (Avellaneda-Stoikov).
Tue	Inventory Skew math.
Wed	Rebate optimization.
Thu	Quote Management (Update freq).
Fri	Live Test (Paper, Illiquid stock).

Institutional Strategy Review

Sharpe >2	Cap Scale	Infra Stable	Plan Q7
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Technical Architecture

- Capacity Analysis (How much capital fits?).
- Infrastructure Cost Benefit Analysis.
- Strategy Correlation Matrix update.
- Team expansion planning.

Autopilot AI Logic

- Which institutional strategies work for retail size?
- Retire HFT if latency is too high.
- Double down on Statistical Arbitrage?
- Review Learning Agent progress.

Operational Playbook

- Document "Institutional Learnings".
- Prepare for "Ecosystem" phase.
- Clean up Research notebooks.
- Upgrade Hardware (Colo?).

Risk & Contingency

- Risk: Complexity. Mitigation: Simplify.
- Risk: Cost. Mitigation: Cut unprofitable infra.
- Risk: Ego. Mitigation: Returns dont lie.

Day-by-Day Schedule

Mon	Performance Deep Dive.
Tue	Infrastructure Audit.
Wed	Strategy Retirement Committee.
Thu	Capacity Planning.
Fri	Q7 Planning.

Licensing & White Labeling Architecture

Tenant Iso	Brand Custom	Lic Key	Rev SaaS
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Technical Architecture

- Multi-Tenancy (Row Level Security).
- White Label UI (CSS Variables).
- License Key Manager (Stripe Integration).
- Admin Super-User Dashboard.

Autopilot AI Logic

- AI Support Agent for external users.
- Tenant Activity Monitoring (Usage metering).
- Proactive "Upsell" triggers.
- Isolate Auto-Pilot instances per tenant.

Operational Playbook

- Separate "Core" IP from "Client" logic.
- Obfuscation of proprietary algos (if on-prem).
- Legal Terms of Service (ToS).
- SLA monitoring.

Risk & Contingency

- Risk: IP Theft. Mitigation: Cloud-only (SaaS).
- Risk: Support burden. Mitigation: AI Chatbot.
- Risk: Cannibalization. Mitigation: Non-compete.

Day-by-Day Schedule

Mon	Multi-Tenant Database Schema.
Tue	Branding/Theme Engine.
Wed	License Verification API.
Thu	Stripe Subscription Integration.
Fri	User Onboarding Flow.

Fund Structure (LP/GP) & Admin

Entity LLC	Docs PPM	Admin Setup	NAV Calc
---------------	-------------	----------------	-------------

Technical Architecture

- Fund Accounting Software integration.
- NAV (Net Asset Value) Calculator.
- LP (Limited Partner) Ledger.
- Fee Calculator (2 & 20).

Autopilot AI Logic

- Calculate High Water Mark per LP.
- Generate "Capital Call" notices.
- Distribute "Quarterly Letter" drafts.
- Reconcile Bank vs Broker vs Ledger.

Operational Playbook

- Legal: Formation of Delaware LP/LLC.
- Admin: Third-party administrator selection.
- Bank: Corporate account setup.
- Audit: Prepare specifically for audit.

Risk & Contingency

- Risk: Compliance (SEC/CFTC). Mitigation: Qualified Counsel.
- Risk: Fraud. Mitigation: Dual signatures.
- Risk: Dispute. Mitigation: Clear PPM.

Day-by-Day Schedule

Mon	Legal Entity Structure Research.
Tue	NAV Calculation Module.
Wed	Fee Engine (Mgmt + Perf).
Thu	Investor Ledger Database.
Fri	PPM (Private Placement Memo) Draft.

Investor Portal & Reporting

Portal Live	Login Secure	K-1 Dist	Trust High
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Technical Architecture

- Secure Client Portal (Separate from Trading App).
- Document Vault (PPM, Subscription Docs, K-1s).
- Performance Charts (Net of Fees).
- 2FA Enforcement.

Autopilot AI Logic

- Generate personalized "Monthly Statement" PDF.
- Answer LP FAQs via Chatbot (RAG on PPM).
- Sentiment analysis of LP emails.
- Alert GP on "Redemption Request".

Operational Playbook

- Data Privacy is paramount.
- Show "Exposure" but hide "Positions" (Protect Alpha).
- Log every view/download.
- Mobile-friendly view.

Risk & Contingency

- Risk: Data Leak. Mitigation: Pen-test portal.
- Risk: Misrepresentation. Mitigation: Disclaimer footers.

Day-by-Day Schedule

Mon	Portal UX Design.
Tue	Auth0 / Cognito Integration.
Wed	Document Vault S3 secure links.
Thu	Performance Charting (Net).
Fri	Security Audit.

Marketing Automation & CRM

Leads Tracked	Email Drip	Meet Booked	AUM Target
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Technical Architecture

- CRM (HubSpot/Salesforce integration).
- Email Automation (SendGrid/Mailgun).
- LinkedIn Scraper (Lead Gen).
- Webinar Host.

Autopilot AI Logic

- Score Leads based on "Net Worth" signals.
- Auto-schedule meetings via Calendly API.
- Drip campaign: "Our Research this Week".
- Personalize outreach using LinkedIn bio.

Operational Playbook

- Segment LPs: Family Office, HNW, Institutional.
- Track "Touchpoints" (Calls/Emails).
- Compliance: Track "General Solicitation" rules (506c vs 506b).
- Analytics on Pitch Deck opens (DocSend).

Risk & Contingency

- Risk: Spamming. Mitigation: Quality over Quantity.
- Risk: Regulatory breach. Mitigation: Accredited checks.
- Risk: Brand damage. Mitigation: Professional tone.

Day-by-Day Schedule

Mon	CRM Schema Setup.
Tue	Lead Scoring Algorithm.
Wed	Email Drip Content.
Thu	LinkedIn Integration.
Fri	Pitch Deck Analytics.

Legal & Compliance (Registration)

Reg Filed	Exam Passed	Man Manual	Off Officer
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Technical Architecture

- Compliance Manual (Digital).
- Trade Allocation Auditor.
- Personal Trading Policy Monitor.
- Archival (Email/Slack/Code).

Autopilot AI Logic

- Pre-Trade Clearance for Employees.
- Scan communication for "Promissory Statements".
- Verify "Best Execution" reports.
- Generate Form ADV updates.

Operational Playbook

- Appoint CCO (Chief Compliance Officer).
- Study for Series 65 (if required).
- Register as CTA (Commodity Trading Advisor) if Futures.
- Cybersecurity policy documentation.

Risk & Contingency

- Risk: Audit Failure. Mitigation: Mock Audit.
- Risk: Fine. Mitigation: Conservative interpretation.
- Risk: Reputation. Mitigation: Clean record.

Day-by-Day Schedule

Mon	Regulatory Map (SEC vs CFTC).
Tue	Compliance Manual Digitization.
Wed	Employee Trade Monitor.
Thu	Archival System (WORM storage).
Fri	Mock Audit.

Hiring First Quant/Dev

Role Defined	Cand Pipeline	Test Hard	Hire 1
-----------------	------------------	--------------	-----------

Technical Architecture

- Code Test Platform (Custom).
- Resume Parser (LLM).
- Onboarding Sandbox (Isolated).
- Access Control (RBAC).

Autopilot AI Logic

- Screen Resumes for "Keywords" (Python, Alpha, Torch).
- Auto-grade Code Tests (Unit tests).
- Generate Interview Questions based on Resume.
- Assess "Cultural Fit" (Async video logic?).

Operational Playbook

- Define Equity/Comp package.
- Protect IP (Non-Compete/IP Assignment).
- Create "Quant Challenge" marketing.
- Interview loop process.

Risk & Contingency

- Risk: Bad Hire. Mitigation: Contract-to-hire.
- Risk: IP Theft. Mitigation: Sandbox.
- Risk: Culture clash. Mitigation: Values alignment.

Day-by-Day Schedule

Mon	Job Description (JD).
Tue	Technical Assessment Design.
Wed	Sourcing Strategy.
Thu	Interview Rubric.
Fri	Sandbox Environment Setup.

Research Grant Program

Grants 3	Ideas 10	Cost Low	IP Owned
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Technical Architecture

- Data Room (Sanitized Dataset).
- Submission Portal (Jupyter Notebooks).
- Evaluation Sandbox.
- Contract Management.

Autopilot AI Logic

- Sanitize Data (Remove PII/Prop signals).
- Evaluate submitted Notebooks for "Overfitting".
- Rank submissions by Sharpe/Novelty.
- Detect Plagiarism.

Operational Playbook

- Partner with Universities.
- Offer "Bounty" for specific problems.
- Clear IP ownership terms.
- Mentorship for top students.

Risk & Contingency

- Risk: Leaking Alpha. Mitigation: Obfuscated data.
- Risk: Low quality. Mitigation: Pre-screen.
- Risk: Legal. Mitigation: Contract structure.

Day-by-Day Schedule

Mon	Data Sanitation Pipeline.
Tue	Legal Grant Agreement.
Wed	Outreach to Universities.
Thu	Evaluation Metric definition.
Fri	Launch "Alpha Grant" page.

Open Source Contribution Strategy

Stars 1k+	PRs Merged	Com Built	Rec Talent
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Technical Architecture

- Open Source Library (e.g., "Apex-Utils").
- Documentation Site.
- Community Discord.
- Contributor License Agreement (CLA).

Autopilot AI Logic

- Auto-triage GitHub Issues.
- Review PRs for style/security.
- Answer community questions.
- Highlight "Contributors" on Leaderboard.

Operational Playbook

- Release non-core components (Connectors, Viz).
- Keep Alpha proprietary.
- Use OS as a recruiting funnel.
- Sponsor key dependencies.

Risk & Contingency

- Risk: Leaking secrets. Mitigation: git-secrets scan.
- Risk: Support drain. Mitigation: "As-is" license.
- Risk: Ford. Mitigation: Maintainership.

Day-by-Day Schedule

Mon	Identify Open Source candidates.
Tue	Repo cleanup & docs.
Wed	License selection (MIT/Apache).
Thu	Publish to PyPI/NPM.
Fri	Announce on Reddit/Hacker News.

Media Strategy (Podcast/Blog)

Subs Growing	Views High	Auth Voice	Guest Big
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Technical Architecture

- Content CMS (Ghost/WordPress).
- Podcast Hosting (Transistor.fm).
- Social Scheduler (Buffer).
- Newsletter Engine (Substack).

Autopilot AI Logic

- Draft Blog Posts from Research Logs.
- Suggest Podcast Interview Questions.
- Clip "Highlights" from audio.
- SEO Optimization.

Operational Playbook

- Weekly "Market Deep Dive".
- Interview other Quants/Traders.
- Transparency focused (Wins & Losses).
- Build the "Personal Brand".

Risk & Contingency

- Risk: Compliance. Mitigation: Disclaimer.
- Risk: Time. Mitigation: Outsourced editing.
- Risk: Haters. Mitigation: Ignore.

Day-by-Day Schedule

Mon	Content Calendar.
Tue	Blog Setup.
Wed	Podcast Equipment Check.
Thu	First Episode Record.
Fri	Launch Strategy.

Conference Speaking & Networking

Talks 3	Net Expanded	Deal Flow	Auth Rec
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Technical Architecture

- Presentation Deck Builder.
- CRM Mobile App.
- Business Card (Digital).
- Follow-up Automation.

Autopilot AI Logic

- Scan Attendee List.
- Prioritize "Must Meet" people.
- Draft "Ice Breaker" research on targets.
- Auto-email follow-up: "Great meeting you..."

Operational Playbook

- Apply to Speak (PyData, QuantCon).
- Host side-events/dinners.
- Sponsor specialized meetups.
- Collect intel on competitors.

Risk & Contingency

- Risk: Travel burnout. Mitigation: Selectivity.
- Risk: COVID/Sick. Mitigation: Hybrid.
- Risk: Bad talk. Mitigation: Rehearse.

Day-by-Day Schedule

Mon	Conference Landscape Map.
Tue	Abstract Submission.
Wed	Deck outline.
Thu	Travel logistics.
Fri	Networking database.

Capital Raise (Series A / Strategic)

Val High	Runway 24mo	Terms Clean	Partner Strat
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Technical Architecture

- Data Room (Financials, Tech, Team).
- Cap Table Simulator (Carta).
- Term Sheet Analyzer.
- Pitch Deck vFinal.

■ Autopilot AI Logic

- Model "Use of Proceeds".
- Simulate "Dilution" scenarios.
- Analyze VC Portfolio fit.
- Track Investor Pipeline velocity.

■■ Operational Playbook

- Roadshow scheduling.
- Due Diligence preparation (Tech/Legal).
- Reference calls.
- Negotiation strategy.

■■ Risk & Contingency

- Risk: Down round. Mitigation: Metrics first.
- Risk: Bad Board member. Mitigation: Reference check the VC.
- Risk: Distraction. Mitigation: Timebox.

■ Day-by-Day Schedule

Mon	Data Room Prep.
Tue	Target Investor List.
Wed	Pitch Rehearsal.
Thu	Financial Model audit.
Fri	Outreach campaign.

M&A; Targets & Acquisition Strategy

Targets 5	Fit Strat	Price Fair	Integ Plan
--------------	--------------	---------------	---------------

Technical Architecture

- Target Screener (Tech Stack match).
- Synergy Calculator.
- Integration Playbook.
- Due Diligence Checklist.

Autopilot AI Logic

- Scan for specific "Niche" competitors.
- Analyze target's GitHub (Code Quality).
- Estimate Target Revenue.
- Model Acquisition ROI.

Operational Playbook

- Buy vs Build decision framework.
- Talent acquisition (Acqui-hire).
- Technology folding.
- Culture merge.

Risk & Contingency

- Risk: Indigestion. Mitigation: Small bites.
- Risk: Culture clash. Mitigation: Leader alignment.
- Risk: Overpayment. Mitigation: Discipline.

Day-by-Day Schedule

Mon	Target Profile definition.
Tue	Screening Process.
Wed	Technical Due Diligence list.
Thu	Synergy Modeling.
Fri	Initial Outreach.

Global Expansion (Geo-Redundancy)

Regions 3	Lat Min	Fail Auto	Comp Local
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Technical Architecture

- Multi-Region AWS/GCP Deployment.
- Global Load Balancer.
- Geo-Replicated Database.
- Compliance (GDPR/Local).

Autopilot AI Logic

- Route traffic to nearest region.
- Shift Load follows the sun (Asia -> EU -> US).
- Failover if Region goes dark.
- Localize UI (Language/Currency).

Operational Playbook

- Establish EU Presence (Dublin/Frankfurt).
- Establish Asia Presence (Singapore/Tokyo).
- Local Legal Counsel.
- Latency reduction for global users.

Risk & Contingency

- Risk: Cost. Mitigation: Reserved Instances.
- Risk: Data Sovereignty. Mitigation: Regional Sharding.
- Risk: Complexity. Mitigation: Terraform.

Day-by-Day Schedule

Mon	Region Selection.
Tue	Infrastructure Replication.
Wed	Data Sovereignty audit.
Thu	Latency Optimization.
Fri	Global Failover Test.

IPO Readiness & Sarbanes-Oxley (SOX)

Audit Clean	Seg Duty	Ctrl Tested	Board Setup
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Technical Architecture

- Internal Controls Framework (COSO).
- Segregation of Duties (Software enforced).
- Whistleblower Hotline.
- Financial Reporting Automation (10-Q ready).

Autopilot AI Logic

- Automated Testing of Internal Controls.
- Detect "Override of Controls".
- Reconcile every penny automatically.
- Generate Board Package.

Operational Playbook

- Appoint Independent Directors.
- Select Auditor (Big 4).
- Draft S-1 Registration Statement (for fun/practice).
- Running the company like a public entity.

Risk & Contingency

- Risk: Bureaucracy. Mitigation: Automate compliance.
- Risk: Cost. Mitigation: Scale revenue first.
- Risk: Distraction. Mitigation: CFO handles it.

Day-by-Day Schedule

Mon	Control Framework Design.
Tue	Segregation of Duties matrix.
Wed	Financial Reporting pipeline.
Thu	Mock Auditor Walkthrough.
Fri	Board Meeting Simulation.

Decentralized Autonomous Organization (DAO)



Technical Architecture

- Governance Token Smart Contract (ERC-20).
- Snapshot Voting Portal.
- Treasury Multisig (Gnosis Safe).
- Proposal Standard (AIP - Apex Improvement Proposal).

Autopilot AI Logic

- Execute "Passed Proposals" automatically on-chain.
- Manage Treasury Diversification.
- Reward "Active Voters".
- Detect "Governance Attack" (Flash loan vote).

Operational Playbook

- Community Management is now Governance.
- Legal Wrapper (DAO LLC vs Foundation).
- Tokenomics Design (Inflation/Burn).
- Airdrop to early contributors?

Risk & Contingency

- Risk: SEC. Mitigation: Utility token only? Or Reg D.
- Risk: Apathy. Mitigation: Incentives.
- Risk: Hack. Mitigation: Timelock.

Day-by-Day Schedule

Mon	Tokenomics Whitepaper.
Tue	Smart Contract Deployment (Testnet).
Wed	Voting Portal Setup.
Thu	Treasury Management Policy.
Fri	Genesis Proposal.

Quantum Computing Integration (Future Proof)

Qubit Sim	Algo Opt	Encr PQC	Speed Exp
--------------	-------------	-------------	--------------

Technical Architecture

- Qiskit / Cirq SDK integration.
- Quantum Annealing (D-Wave) for Portfolio Opt.
- Post-Quantum Cryptography (kyber).
- Quantum Monte Carlo.

Autopilot AI Logic

- Solve "Knapsack Problem" (Portfolio Alloc) on QPU.
- Quadratic Unconstrained Binary Optimization (QUBO).
- Encrypt keys with PQC standards.
- Simulate Quantum Advantage.

Operational Playbook

- Partner with IBM/Google Cloud Quantum.
- Hybrid Classical/Quantum workflow.
- Research: "Quantum Machine Learning".
- Hype management.

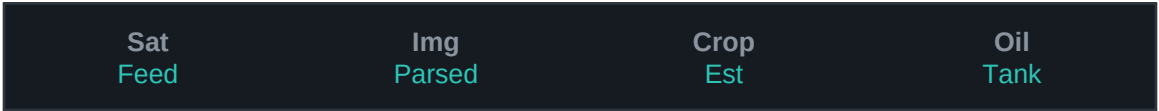
Risk & Contingency

- Risk: It's too early. Mitigation: Research only.
- Risk: Cost. Mitigation: Simulation.
- Risk: Talent. Mitigation: PhD consultants.

Day-by-Day Schedule

Mon	Qiskit Environment Setup.
Tue	Portfolio Optimization as QUBO.
Wed	Quantum Monte Carlo prototype.
Thu	Post-Quantum Crypto migration plan.
Fri	Run job on real Quantum Hardware.

Space/Satellite Data (Orbital Alpha)



Technical Architecture

- Satellite Imagery Provider (Planet/Maxar).
- Computer Vision Model (Counting Objects).
- Geolocation Mapping.
- Shadow Analysis (Oil tank levels).

Autopilot AI Logic

- Count cars at Walmart -> Predict Earnings.
- Measure "Shadow Length" on Oil Tanks -> Inventory.
- Monitor Crop Health (NDVI) -> Ag Futures.
- Detect Ship movements (Supply Chain).

Operational Playbook

- Expensive data. Cost/Benefit analysis.
- Latency is high (daily/weekly).
- Cloud storage for massive imagery.
- Correlate with "Alternative Data".

Risk & Contingency

- Risk: Clouds blocking view. Mitigation: Radar (SAR).
- Risk: False correlation. Mitigation: Ground truth.
- Risk: Privacy. Mitigation: Aggregated data.

Day-by-Day Schedule

Mon	Satellite API Access.
Tue	Computer Vision Model (YOLO/ResNet).
Wed	Retail Parking Lot Strategy.
Thu	Oil Inventory Strategy.
Fri	Agricultural Yield Strategy.

Philanthropic Algo (Effective Altruism)

Give 10%	Impact High	Tax Ded	Soul Good
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Technical Architecture

- Donation API (Every.org / GiveWell).
- Impact Calculator.
- Donor Advised Fund (DAF) API.
- Tax Optimization (Donating appreciated stock).

Autopilot AI Logic

- If Week P&L; > Target -> Sweep 10% to Charity.
- Identify "Most Effective" charities (Lives Saved/\$).
- Donate "Winners" directly to avoid Capital Gains tax.
- Report on "Social Impact" alongside Financials.

Operational Playbook

- Set up DAF.
- Define "Giving Charter".
- Engage employees/community in selection.
- Public transparency (if desired).

Risk & Contingency

- Risk: Giving away too much. Mitigation: Cap.
- Risk: Scam charity. Mitigation: Vetted list.
- Risk: Tax audit. Mitigation: Receipts.

Day-by-Day Schedule

Mon	Charity Selection Logic.
Tue	Donation API integration.
Wed	Tax-Efficient Donation module.
Thu	Impact Dashboard.
Fri	The First Automated Grant.

Succession Planning & Knowledge Transfer

Bus Factor	Docs Full	Key Man	Fail Safe
---------------	--------------	------------	--------------

Technical Architecture

- Dead Man's Switch.
- Knowledge Base (Wiki/Video).
- Access Recovery Protocol.
- Legal Will / Trust specific to Digital Assets.

Autopilot AI Logic

- If Admin inactive for 30 days -> Trigger "Safe Mode".
- Notify "Trusted Contacts" with access keys.
- Auto-liquidate risky positions?
- Transfer ownership of DAO keys.

Operational Playbook

- Train a "Deputy".
- Simplify the system (Reduce "Bus Factor" of 1).
- Record "Masterclass" videos explaining the core.
- Legal review of Digital Asset inheritance.

Risk & Contingency

- Risk: Keys lost forever. Mitigation: Shamir's Secret Sharing.
- Risk: Rogue Deputy. Mitigation: Multisig.
- Risk: Obsolescence. Mitigation: Maintenance fund.

Day-by-Day Schedule

Mon	Dead Man's Switch logic.
Tue	Key Recovery Ceremony.
Wed	Succession Documentation.
Thu	System Simplification audit.
Fri	"Fire Drill" (Simulate absence).

The "Black Box" (Unsupervised Learning)

Cluster New	Feature Auto	Ret Uncorr	Explain Low
----------------	-----------------	---------------	----------------

Technical Architecture

- Autoencoder for Anomaly Detection.
- Clustering (DBSCAN/K-Means) on Price Action.
- Generative Adversarial Nets (GAN) for Data Augmentation.
- Self-Supervised Learning.

Autopilot AI Logic

- AI segments market regimes without labels.
- Generates synthetic data to train itself.
- Finds "Hidden States" (HMM).
- Trades patterns that "look weird" but work.

Operational Playbook

- Hard to explain to investors.
- Requires massive data.
- Strict risk limits (because we don't know "Why").
- Monitor Feature Importance shifts.

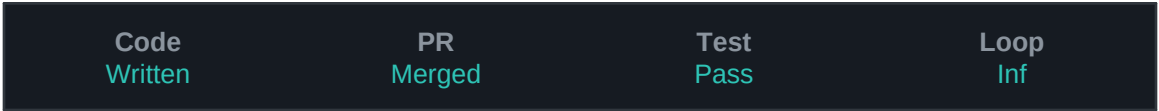
Risk & Contingency

- Risk: Model drift. Mitigation: Re-train daily.
- Risk: Black Swan. Mitigation: Hard stops.
- Risk: Regulation (Explainable AI). Mitigation: Parallel interpretability.

Day-by-Day Schedule

Mon	Autoencoder Architecture.
Tue	Clustering Analysis.
Wed	GAN for Synthetic Data.
Thu	Hidden Markov Models.
Fri	Unsupervised Strategy Launch.

System Singularity (Self-Coding)



Technical Architecture

- LLM Coder (Fine-tuned on Repo).
- CI/CD with "Self-Correction".
- Sandbox Execution Environment.
- Codebase Indexer.

Autopilot AI Logic

- Identify Bug -> Write Fix -> Run Test -> Merge.
- Identify Efficiency -> Refactor Code -> Benchmark -> Merge.
- Create new "Tool" for itself (e.g., "I need a better scraper").
- The loop: Observe, Orient, Decide, Act, Code.

Operational Playbook

- Human MUST approve PRs (for now).
- Limit scope to "Non-Critical" systems first.
- Version Control is the safety net.
- Monitor "Cyclomatic Complexity".

Risk & Contingency

- Risk: Infinite Loop. Mitigation: Timeout.
- Risk: Deleting production. Mitigation: Permissions.
- Risk: Skynet. Mitigation: Unplug it.

Day-by-Day Schedule

Mon	Repo Indexing & Embeddings.
Tue	"Issue to PR" Agent.
Wed	Test-Driven Development Agent.
Thu	Refactoring Agent.
Fri	The First Self-Merged PR.

The Century Portfolio (100-Year View)

Horizon 100y	Asset Real	Inf Hedge	Survive Tags
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Technical Architecture

- Lindylania (Things that have survived long will survive longer).
- Geo-Political Hedge.
- Hard Assets (Gold, Land, Bitcoin).
- Dynasty Trust Structure.

Autopilot AI Logic

- Rebalance only once per year.
- Ignore "News". Ignore "Cycles".
- Focus on "Survival" not "Maximization".
- Buy "Farming Land", "Water Rights", "Energy Consumers".

Operational Playbook

- Low turnover. Extremely low cost.
- Custody is the main risk.
- Diversification across Jurisdictions.
- The "Do Nothing" portfolio.

Risk & Contingency

- Risk: Confiscation. Mitigation: Decentralization.
- Risk: War. Mitigation: Neutral countries.
- Risk: Technology shift. Mitigation: Indexing.

Day-by-Day Schedule

Mon	100-Year Asset Selection.
Tue	Inflation/Deflation Resilience.
Wed	Jurisdiction Diversification.
Thu	Custody/Trust Setup.
Fri	The Purchase.

Global Macro Hedge Fund Launch

Launch Day	AUM \$100M+	Global Trade	Legacy Start
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Technical Architecture

- institutional Prime Broker (Goldman/Morgan Stanley).
- ISDA Agreements.
- Global Trading Desk (24/7 staffed).
- Bloomberg Terminal bloat.

Autopilot AI Logic

- The AI is now the "CIO" support.
- Human Traders execute the "Art". AI executes the "Science".
- Global Risk Book monitor.
- Macro Regime detection across 50 countries.

Operational Playbook

- Hire "Head of Trading", "CRO", "CFO".
- Office Space (or Remote First?).
- Compliance, Compliance, Compliance.
- Investor Relations roadshow.

Risk & Contingency

- Risk: Scale breaks alpha. Mitigation: Cap AUM.
- Risk: Culture dilution. Mitigation: Hiring bar.
- Risk: Stress. Mitigation: Meditation.

Day-by-Day Schedule

Mon	Prime Broker Selection.
Tue	ISDA Negotiation.
Wed	Hiring the C-Suite.
Thu	Global Macro Strategy Definition.
Fri	"Open for Business".

Writing the Book (Memoir/Technical)

Pages 300	Pub Wiley	Best Seller	Code Shared
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Technical Architecture

- Book Outliner (LLM).
- Code Snippet Extractor.
- Literary Agent.
- Publisher Pitch.

Autopilot AI Logic

- Analyze 2 Years of Logs to find "Stories".
- Draft Chapters based on "Weekly Updates".
- Visualize the "Equity Curve" journey.
- Fact Check dates and trades.

Operational Playbook

- Write 1000 words/day.
- Decide: Self-Publish vs Traditional.
- Marketing Plan.
- The "Technical Companion" website.

Risk & Contingency

- Risk: Boreodm. Mitigation: Tell stories, not just code.
- Risk: Secrets. Mitigation: Redact alpha.
- Risk: Time. Mitigation: Ghostwriter?

Day-by-Day Schedule

Mon	Outline & Chapter Structure.
Tue	Data Mining for Stories.
Wed	Drafting "The Beginning".
Thu	Drafting "The Crisis".
Fri	Drafting "The Success".

The Final Optimizations

Code Perfect	Perf Peak	Docs Gold	Ready Exit
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Technical Architecture

- Code Freeze.
- Full System Audit.
- Performance Profiling (Flame graphs).
- UI Polish (Pixel perfect).

Autopilot AI Logic

- Run the "Grand Regression Test".
- Verify every single strategy one last time.
- Clean up database (Vacuum).
- Optimize cloud costs final pass.

Operational Playbook

- Prepare the "Golden Master" release.
- Update Copyright dates.
- Finalize all legal paperwork.
- Team "Thank You" bonuses.

Risk & Contingency

- Risk: Breaking change at the end. Mitigation: Freeze.
- Risk: Over-polishing. Mitigation: Ship it.

Day-by-Day Schedule

Mon	Code Freeze Enforced.
Tue	Regression Testing.
Wed	Performance Tuning.
Thu	UI/UX Final Review.
Fri	The Golden Build.

Mission Accomplished. The 2-Year Journey Ends.

Done
Yes

Proud
Yes

Next
Plan

Champagne
Pop

Technical Architecture

- The "Shutdown" protocol (or "Transfer").
- Legacy Archive.
- The Next Master Plan (V4?).
- Vacation Mode.

Autopilot AI Logic

- AI sends "Congratulations" email.
- Generates "Highlight Reel" video.
- Calculates Total 2-Year Returns.
- Switches to "Maintenance Mode".

Operational Playbook

- Closing Dinner.
- Reflect on the journey.
- Update LinkedIn.
- Sleep.

Risk & Contingency

- Risk: None. You made it.
- Risk: Withdrawal (from coding). Mitigation: New Project.
- Risk: Purpose void. Mitigation: Philanthropy.

Day-by-Day Schedule

Mon	Final System Check.
Tue	Team Retrospective.
Wed	Archival Process.
Thu	The Last Commit.
Fri	CELEBRATION.